

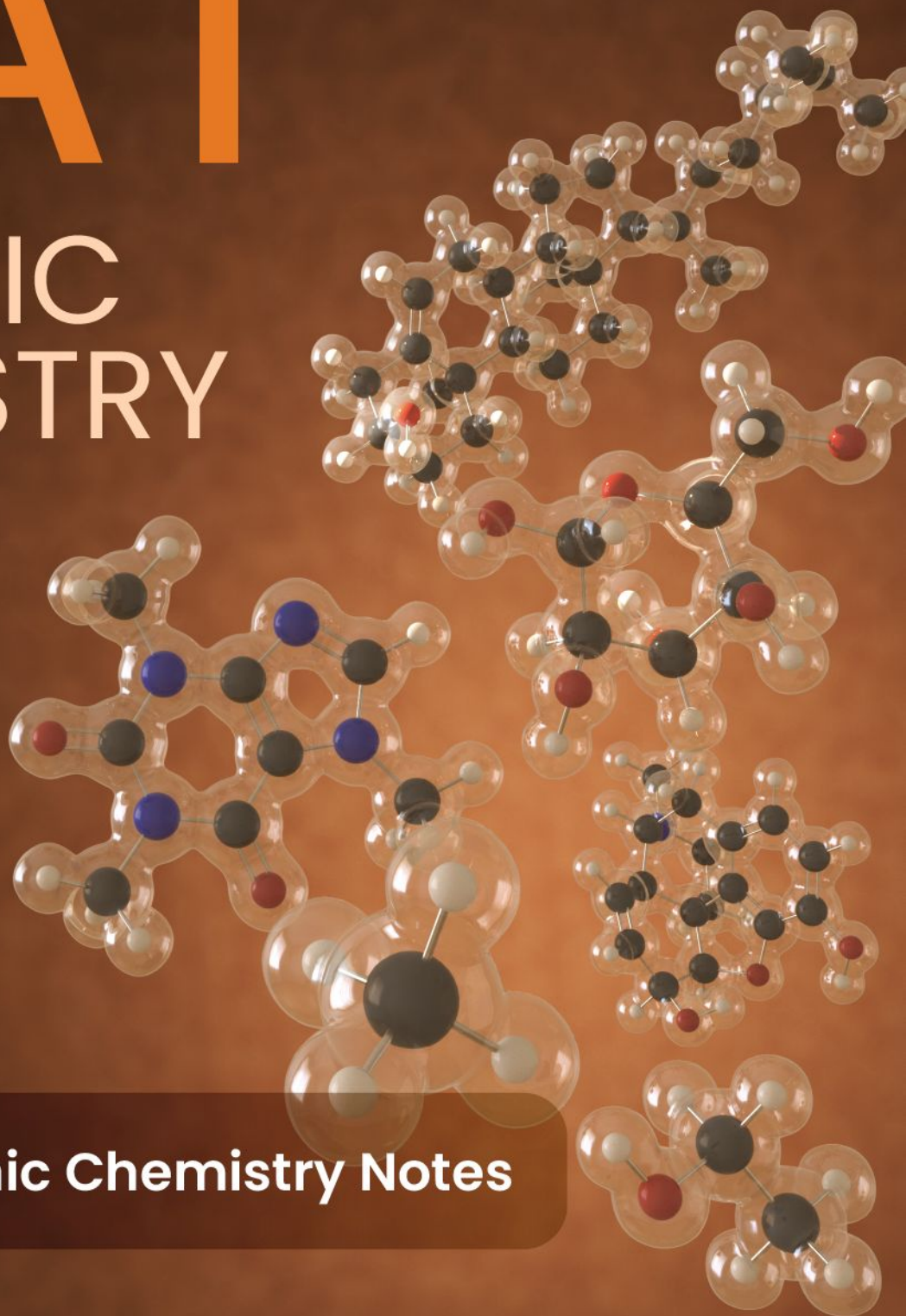
DAT

ORGANIC CHEMISTRY

2026 EDITION

Everything you
need to know
to get a great
DAT organic
chemistry score

Dr. Mike's Organic Chemistry Notes



Ch. 1: Bonding, Resonance, and Molecular Geometry

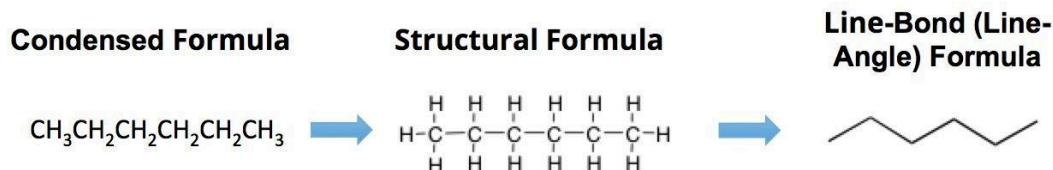
Lesson 1 – Meet Dr. Mike Christiansen!

👋 -Dr. Mike

Lesson 2 – Molecular Bonding Geometry and Hybridization

Electron Domains	Geometry	Bond Angle	Hybridization
2	Linear	180	sp
3	Trigonal planar	120	sp ²
4	Tetrahedral	109.5	sp ³

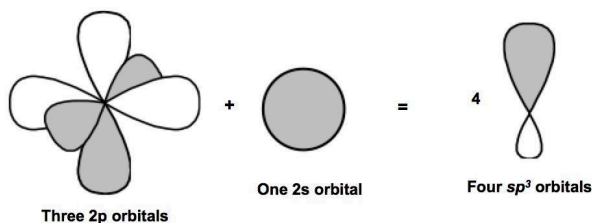
Lesson 3 – Condensed Formulas and Line-Bond Formulas



Lesson 4 – Sigma and Pi Bonds

Type of covalent bond	Number of bonds
Single bond	one sigma (σ)
Double bond	one sigma (σ) + one pi (π)
Triple bond	one sigma (σ) and two pi (π)

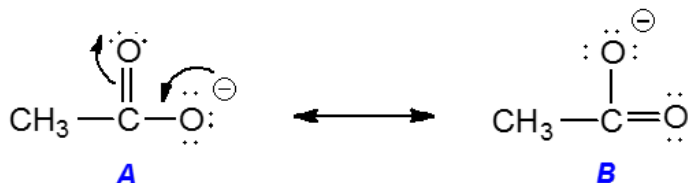
Lesson 5 – Orbital Hybridization



Carbon's electron configuration: $1s^2 2s^2 2p^2$

Lesson 6 – Resonance StructuresDefinition

There are some molecules that have *pi electrons* that can move around from one atom to another. For example, the following molecules (A and B) are both different forms of *acetate*:



Structures A and B are called **resonance structures** (or **resonance contributors**). In reality, **acetate** actually exists somewhere in-between A and B, with the – charge being shared equally by the two oxygens.

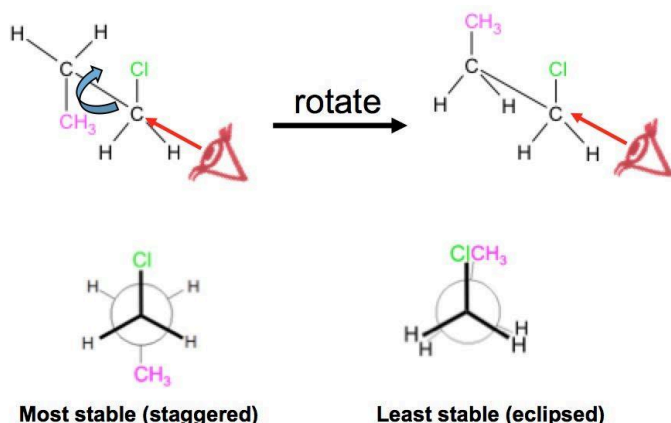
Resonance Rules

When drawing different **resonance structures**, remember:

1. **Only electrons move.** Specifically, only **pi electrons**, lone-pair electrons, or negative charges can move. Do NOT move atoms.
2. You CAN move electrons toward or into an atom that does NOT have a full octet, such as **carbocations**.
3. If an atom already HAS a full octet, then you can move electrons into it ONLY IF you push electrons out the opposite side (electrons in, electrons out).
4. Do not move or break **sigma bonds**, only **pi bonds**.

Determining Greatest Resonance Contributor

1. The most stable **resonance structure** will have a full octet on every atom.
2. The most stable **resonance structure** will have the smallest possible number of charges.
3. The most stable **resonance structure** will have negative charges on the most electronegative atoms and positive charges on the least electronegative atoms.

Lesson 7 – Newman Projections

Order of **stability** in Newman Projections
from most to least:

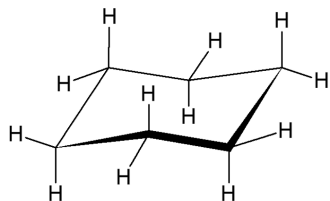
Staggered > Gauche > Eclipsed

(most stable) <-----> (least stable)

Lesson 8 – Cycloalkanes and Ring Strain

Cycloalkanes are alkanes that are **cyclic** –in other words, **ringed alkanes**, or alkanes with rings in them. Cyclohexane is the most stable cycloalkane.

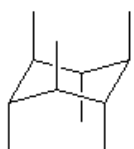
How to draw **chair conformations** of cyclohexane (follow along!):



Axial vs. Equatorial



equatorial bonds



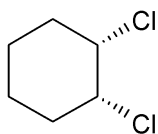
axial bonds

- **Equatorial** positions are more **stable** (lower energy) than axial positions for larger groups because of 1,3-diaxial interactions.
- Placing the **largest** substituents in the equatorial positions will usually achieve the greatest stability in cyclohexane rings.

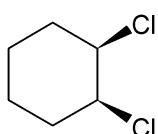
Trans vs. Cis cyclohexanes

Cis – two substituents going in **same** direction

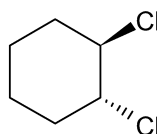
Trans – two substituents going in **opposite** directions



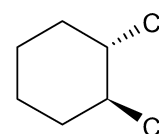
cis-1,2-dichloro-
cyclohexane



cis-1,2-dichloro-
cyclohexane



trans-1,2-dichloro-
cyclohexane

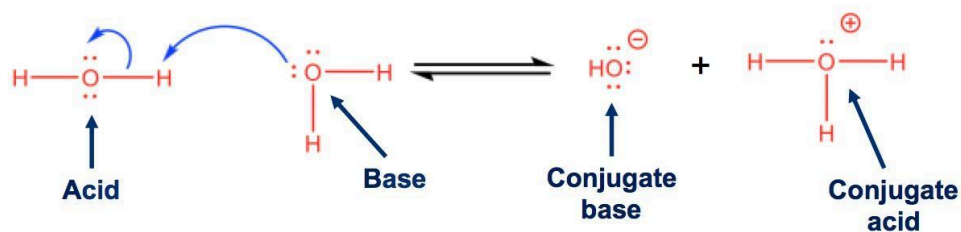


trans-1,2-dichloro-
cyclohexane

Ch. 2: Acids and Bases

Lesson 1 – Acid-Base Definitions

- A **Lewis acid** is a substance that accepts electrons.
- A **Lewis base** is a substance that donates electrons.



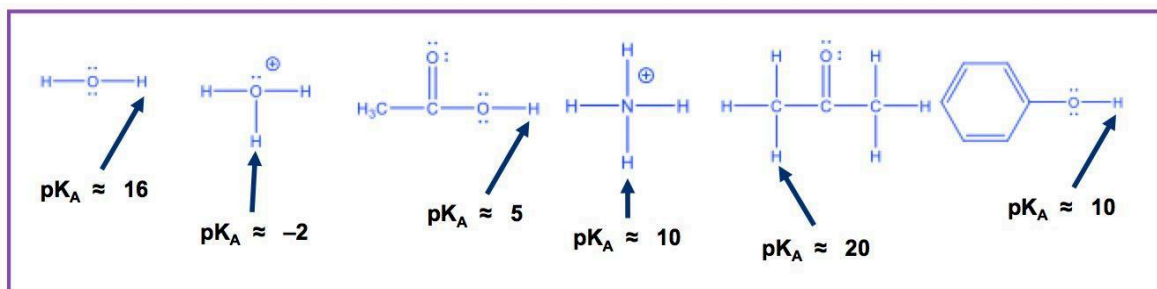
Lesson 2 – Conjugate Base-Acid Relationship and pH Scale

- The stronger the **acid**, the weaker its **conjugate base**.
- The stronger the **base**, the weaker its **conjugate acid**.

$$\uparrow K_A = \downarrow pK_A = \uparrow \text{acid strength}$$

- The more stable/weaker the **conjugate base**, the stronger the **acid**.
- The more stable/weaker the **conjugate acid**, the stronger the **base**.

pK_as for Organic Compounds



Lesson 3 – Ranking Acids and Bases with CARDIO (Charge)

If all other factors are the same (or close to the same), then:

- The more **positively-charged** the compound = the **more acidic**
- The more **negatively-charged** the compound = the **more basic**

Lesson 4 – Ranking Acids and Bases with CARDIO (Atom)

If all other factors are about the same, then hydrogen's acidity **increases** as the atom that it's bonded to:

- goes **left-to-right across a row** on the periodic table (increasing electronegativity)
- goes **down a column** on the periodic table (increasing size)

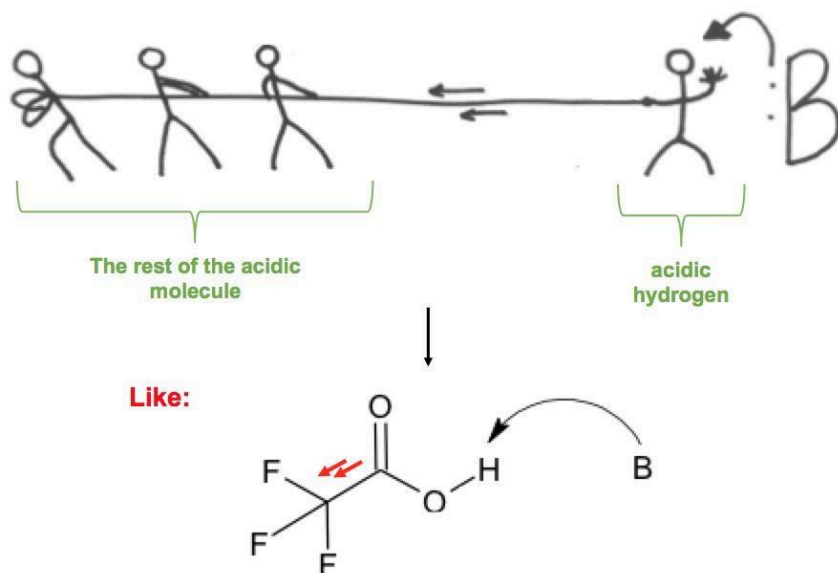
Lesson 5 – Ranking Acids and Bases with CARDIO (Resonance)

- The more stable the **conjugate base**, the stronger the **acid**
- The more stable the **conjugate acid**, the stronger the **base**

Resonance increases the stability of charges, therefore a **resonance-stabilized conjugate base** will be a **stronger acid**.

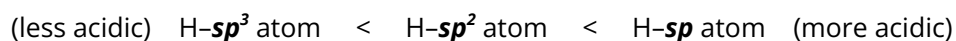
Lesson 6 – Ranking Acids and Bases with CARDIO (Dipole Induction)

- **Electron Withdrawing** groups **increase** acidity
- **Electron Donating** groups **decrease** acidity



Lesson 7 – Ranking Acids and Bases with CARDIO (Orbitals)

If all other factors are about the same, then acidity follows the below trend:



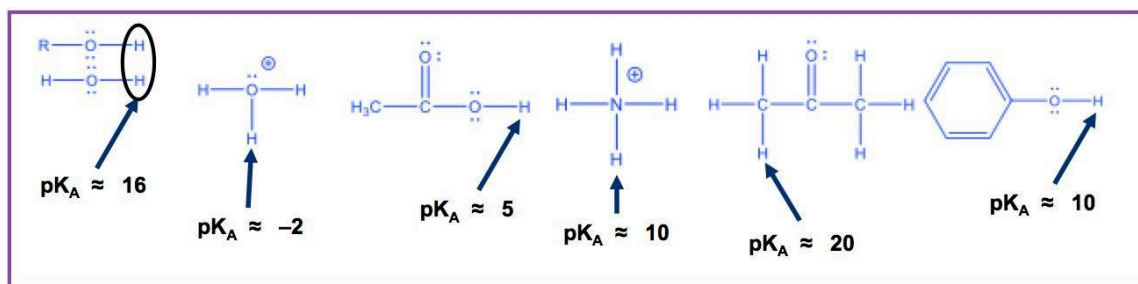
- **S-orbitals** tend to be more electronegative, so the more “s-character” an atom has, the stronger the acid.

Lesson 8 – Acid and Base Review

How to Sort Acids and Bases by Strength

First, convert the acid to its conjugate base.

- 1. Charge** – Positively charged compounds are typically more acidic, negatively charged compounds are typically more basic.
- 2. Atom** – The more electronegative/larger the atom with a negative charge, the more acidic the hydrogen is.
- 3. Resonance** – The more resonance-stabilized the conjugate base, the stronger the acid.
- 4. Dipole Induction** – Electron withdrawing groups increase acidity, electron donating groups decrease acidity.
- 5. Orbitals** – The more s-character an atom has, the more electronegative it is, and the more acidic hydrogens bonded to it will be. i.e. $sp^3 < sp^2 < sp$



Lesson 9 – pH and Amino Acids

To determine the structure or charge of any amino acid:

- If the **pH of the solution is LOWER than the pKa** of the functional group, the functional group will be *protonated*.
- If the **pH of the solution is HIGHER than the pKa** of the functional group, the functional group will be *deprotonated*.

Functional Group	pH < pK _a (protonated)	pH > pK _a (deprotonated)
amino group	-NH ₃ ⁺	-NH ₂
carboxylic acid group	-COOH	-COO ⁻
hydroxyl group	-OH	-O ⁻

Ch. 3: Nomenclature

Lesson 1 – IUPAC Basics and Naming Alkanes

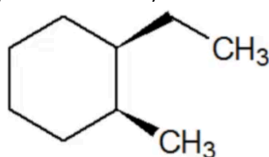
Naming Alkanes

1. Find the parent chain, the longest carbon chain. If two possible parent chains have the same length, but different substituent numberings, pick the one with the smaller substituent number at the first point of difference.
2. Count the number of carbon atoms in the **parent chain** and match that number to the name from our earlier chart (methane, ethane, propane, etc.)
3. Identify and number the **substituents** (appendage dangling off the **parent chain**) in whichever direction gives them the lowest number.
4. Write the name as a single word with the substituents in alphabetical order.

Lesson 2 – Naming Cycloalkanes and Alkyl Halides

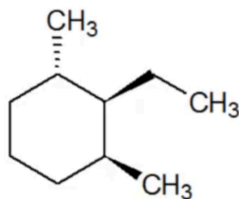
Naming Cycloalkanes

1. When there are two substituents on a cyclic molecule, their direction must be indicated with prefix “cis”, meaning “same side”, or “trans”, meaning “opposite sides”.



cis-1-ethyl-2-methylcyclohexane

2. If there are more than two substituents, “cis” and “trans” are no longer enough, and these substituents must be named with their stereochemical configurations (R/S system).



(1S,3S)-2-ethyl-1,3-dimethylcyclohexane

Naming Alkyl Halides

When naming alkyl halides, we follow the same rules for naming alkanes, except that we use prefixes “fluoro-”, “chloro-”, “bromo-”, or “iodo-”, to identify each halogen substituent. Alternatively, we may use suffixes “-yl fluoride”, “-yl chloride”, “-yl bromide”, or “-yl iodide”.

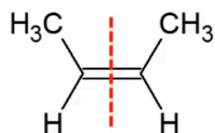


1-chlorohexane
or hexyl chloride

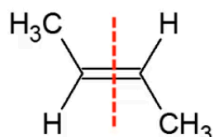
Lesson 3 - Naming Alkenes and Alkynes

Naming Alkenes

1. Use the suffix "**-ene**" instead of "**-ane**".
2. Add a number at the start of the double bond.
3. Add prefixes "*cis*" or "*trans*" for alkenes with different priority substituents, and at least one hydrogen on either side of the alkene.
4. If all substituents are different, use the E/Z naming system, where E = highest priority substituents on opposite sides, and Z = highest priority substituents on the same side.



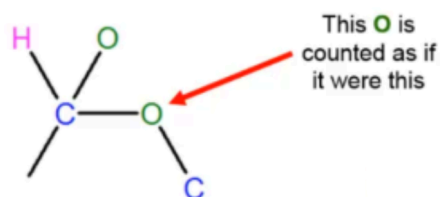
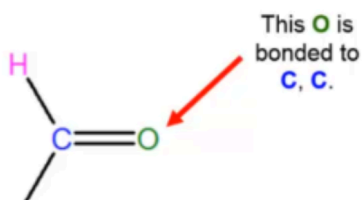
cis-2-butene or
cis-but-2-ene



trans-2-butene or
trans-but-2-ene

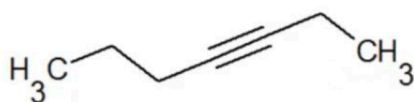
Determining Priority

1. Highest atomic number = highest priority.
2. If there's a tie, keep going to adjacent carbons until you break the tie.
3. Multiple-bonded atoms are counted as the same number of single-bonded atoms:



Naming Alkynes

1. Use the suffix "**-yne**" instead of "**-ane**".
2. Add a number at the start of the triple bond.



hept-3-yne

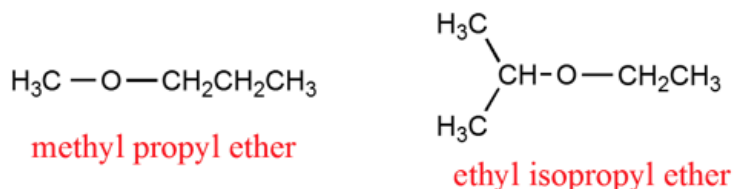
Lesson 4 – Naming Alcohols, Ethers, and Amines

When naming **alcohols**, we follow the rules for naming alkanes, except:

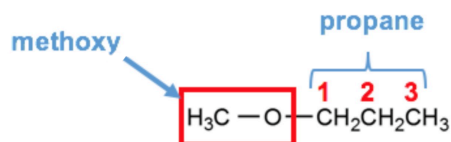
1. The parent chain is now the longest chain that has the hydroxyl group, even if there are longer carbon chains available.
2. Number the carbon chain in the direction that gives the smallest number to the carbon bonded to the hydroxyl group.
3. Hydroxyl groups are higher priority than cycloalkanes, amines, alkenes, ethers, and alkyl halides, so they must be numbered according to the lowest-number carbon that is bonded to the hydroxyl group.
4. Change the suffix “-e” to “-ol”.

When naming **ethers**:

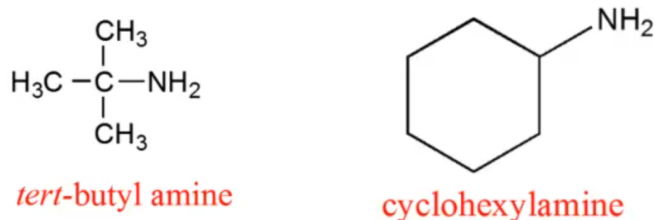
1. Name the two alkyl groups as substituents with “ether” at the end:



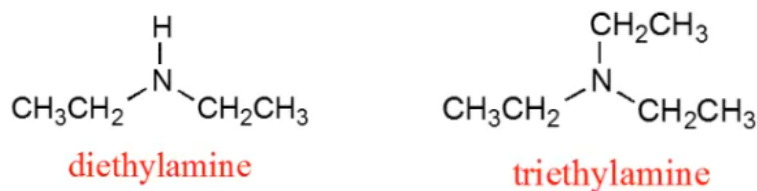
2. Consider the longest carbon chain to be the parent chain and the alkoxy group to be a substituent:



When naming **primary amines**, add the suffix “amine” to the name of the organic substituent.



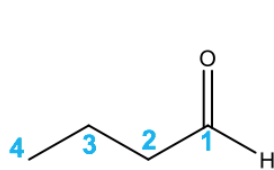
Symmetrical and **secondary amines** are named by adding “di-” or “tri-” to the alkyl group:



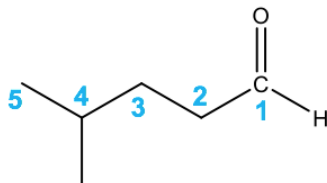
Lesson 5 - Naming Aldehydes and Ketones

When naming **aldehydes**, we follow the rules for naming alkanes with the addition of two rules:

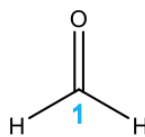
1. We number the parent chain in the direction that gives highest priority (lowest number) to the aldehyde (carbonyl) carbon.
2. We replace "e" with "al".



butanal



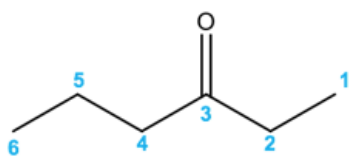
4-methylpentanal



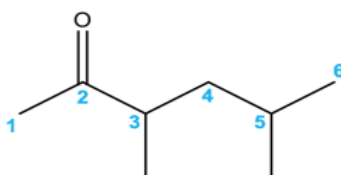
methanal
(formaldehyde)

When naming **ketones**, we follow the same rules for naming alkanes, except:

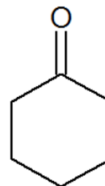
1. Similar to aldehydes, the parent chain must be chosen with priority given to the ketone (carbonyl) carbon.
2. Replace the "e" with "one".
3. The carbonyl carbon in a cyclic ketone is assumed to be the #1 carbon.



3-hexanone



3,5-dimethyl-2-hexanone
or 3,5-dimethylhexan-2-one

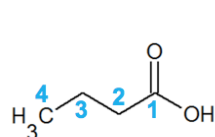


cyclohexanone

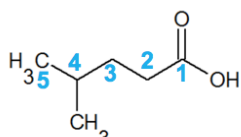
Lesson 6 – Naming Carboxylic Acids and Derivatives

When naming **carboxylic acids**, we follow the rules for naming alkanes, except:

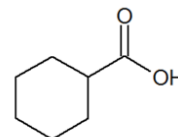
1. We number the parent chain in the direction that gives the highest priority (lowest number) to the carboxylic acid group.
2. We replace “e” for “**oic acid**”.



butanoic acid



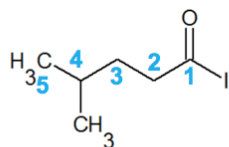
4-methylpentanoic acid

methanoic acid
(formic acid)

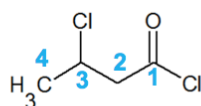
cyclohexanecarboxylic acid

When naming **acid halides**:

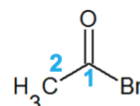
1. Follow the same rules as for carboxylic acids, and change the suffix to “**oyl halide**”.



4-methylpentanoyl iodide

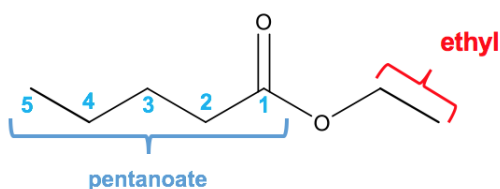


3-chlorobutanoyl chloride

ethanoyl bromide
(acetyl bromide)

When naming **esters**:

1. The alkyl group attached to the ester oxygen gets listed first with the suffix “**yl**”. The parent chain then follows.
2. The parent chain starts at the carbonyl carbon and is counted moving away from the ester oxygen. Parent chain's suffix is replaced with “**oate**”.

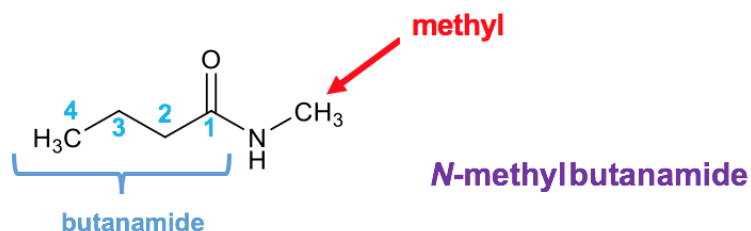


ethyl pentanoate

Lesson 6 – Naming Carboxylic Acids and Derivatives (Continued)

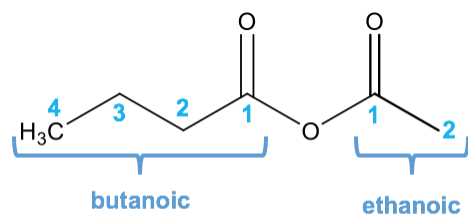
When naming **amides**, we follow the same pattern of naming for **esters**, except:

1. Any alkyl groups attached to the nitrogen gets listed as "N-methyl", "N-ethyl", "N-propyl" etc.
2. The parent chain starts at the carbonyl carbon and is counted moving away from the amide nitrogen.
3. In the name, the **parent chain's** ending "**e**" is replaced with "**amide**".



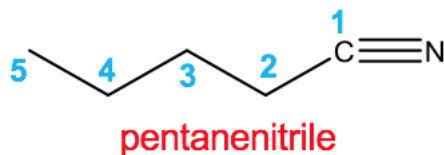
When naming **acid anhydrides**:

1. Determine the length of the chain on either side of the bridging oxygen.
2. List both lengths alphabetically, replacing each suffix "**e**" with "**oic**".
3. Write "**anhydride**" at the end of the name.



When naming **nitriles**, follow the same rules for naming alkanes, except:

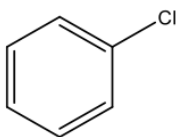
1. The parent chain is the longest carbon chain that involves the nitrile carbon.
2. Number the parent chain in the direction that gives the smallest number to the nitrile carbon.
3. Add the suffix "nitrile" to the parent name.



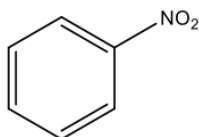
Lesson 7 – Naming Aromatics

When naming **substituted benzenes**:

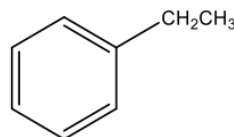
1. Identify the **parent chain**, which is the benzene containing the highest-priority functional group. That **parent chain** is the **parent chain name**.
2. The carbon atom in the ring that is attached to the priority functional group is numbered as carbon #1.
3. Number around the ring in whichever direction (clockwise or counterclockwise) that gives the lowest number at the first point of difference.
4. If the numbers are the same in both directions, pick the one that gives the lower number to the substituent that is alphabetically first.



chlorobenzene



nitrobenzene



ethylbenzene

Lesson 8 – Naming Polyfunctional Compounds

When naming **poly-functional compounds**:

1. You must identify the highest-priority functional group. The **parent chain** containing this functional group is the **parent chain name**.
2. Once you identify the **parent chain** and its functional group, follow the naming rules for that particular functional group. All other functional groups in the molecule are considered and named as substituents.

Priority of Functional Groups

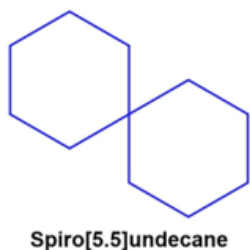
Functional group (general formula) R = any C _x H _y chain	Examples (if parent):	If it's a substituent to a higher-priority group, then it's called:
Carboxylic acid (RCO ₂ H)	Ethanoic acid, propanoic acid, butanoic acid, etc.	
Ester (RCO ₂ R')	Ethyl ethanoate, methyl propanoate, etc.	Alkoxycarbonyl
Acid chloride (RCOCl)	Ethanoyl chloride, propanoyl chloride	Chloroformyl
Amide [R'C(O)NR ₂]	N,N-dimethyl propanamide; N-ethylbutanamide	Amido
Nitrile (R-C≡N)	Ethanenitrile, propanenitrile, etc.	Cyano
Aldehyde ()	Ethanal, propanal, butanal, etc.	Oxo or formyl
Ketone ()	2-butanone, 3-butanone, 2-pentanone	Oxo
Alcohol (ROH)	Methanol, ethanol, propanol, etc.	Hydroxy
Phenol (PhOH)		
Amine (R'-NR ₂)	N,N-dimethylpropanamine, N-ethylcyclohexanamine	Amino
Alkene ()	Ethene, propene, butene, etc.	
Alkyne ()	Ethyne, propyne, butyne, etc.	
Ether (R'OR)	Methoxyethane, ethoxypropane	methoxy, ethoxy, propoxy, etc.
Alkyl halide (RX) X = F, Cl, Br, or I	2-bromopropane, 3-chlorobutane, etc.	Fluoro, chloro, bromo, iodo, etc.
Nitro (NO ₂)	N/A	Nitro

Lesson 9 - Naming Spiro and Bicyclic Alkanes

Spiro Alkanes

IUPAC names for **spiro alkanes** have the format **spiro[a.b]parent name**.

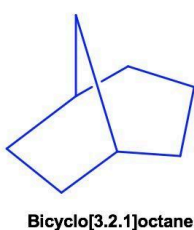
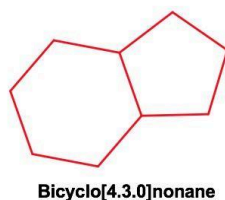
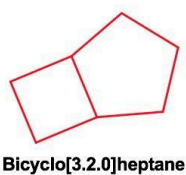
1. Count the **total** number of carbons across the entire molecule. This tells you the alkane **parent name** that goes at the end.
2. Count the number of carbons to the left, and to the right of your spiro-carbon center. These numbers are **a** and **b** in your IUPAC name, listed from lowest to highest.
3. Write your final IUPAC name as **spiro[a.b]parent name**.



Bicyclic Alkanes

IUPAC names for **bicyclic alkanes** have the format **bicyclo[a.b.c]parent name**.

1. Count the **total** number of carbons across the entire molecule. This tells you the alkane **parent name** that goes at the end.
2. Count the number of carbons to the left, and to the right, and above your **bridgehead carbons**. These numbers are a, b, and c in your IUPAC name, listed from highest to lowest.
3. Write your final IUPAC name as bicyclo[a.b.c]parent name.

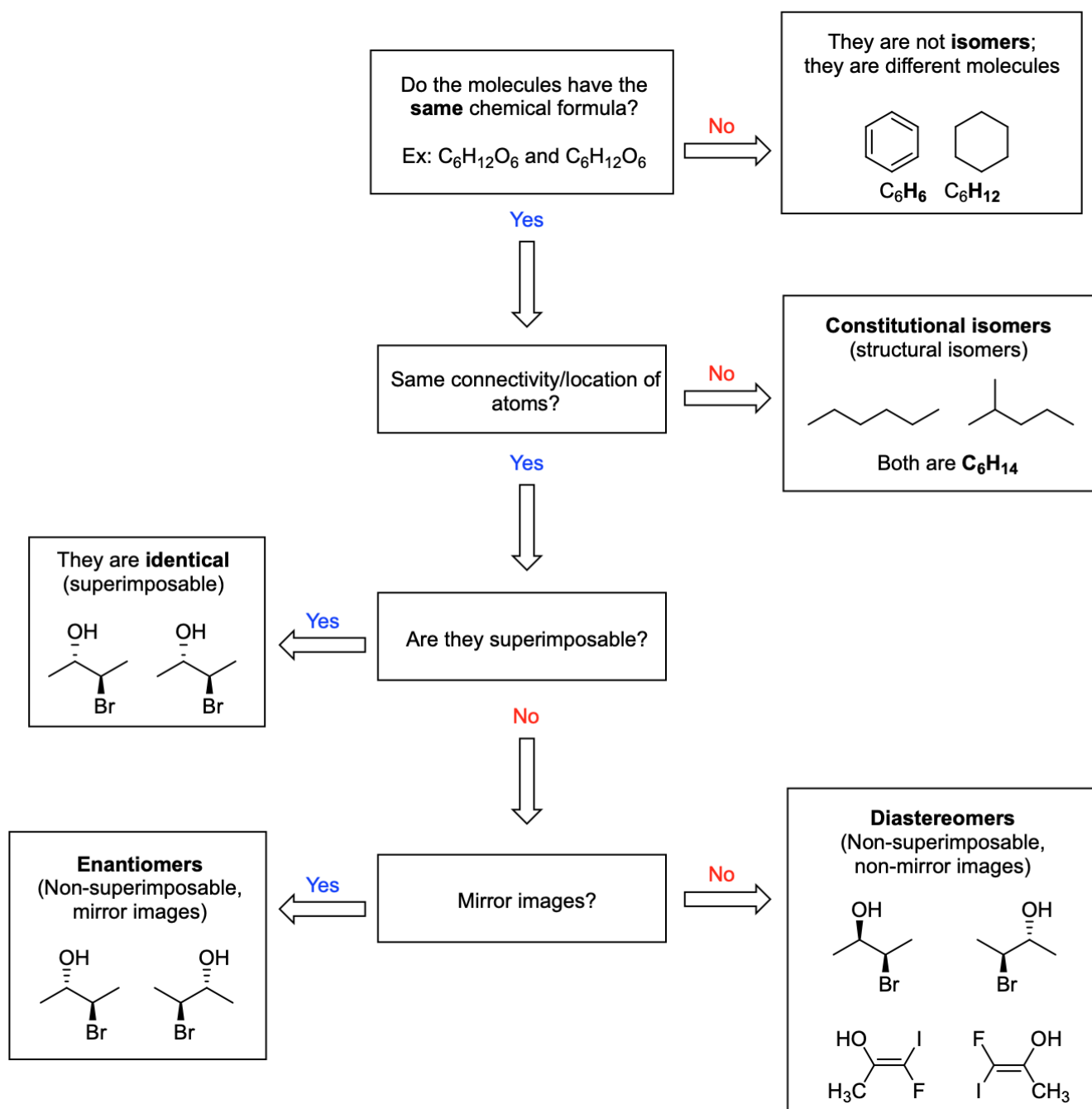


Ch. 4: Stereochemistry

Lesson 1 - Isomers

Constitutional Isomers vs. Diastereomers vs. Enantiomers

- Molecules that have the same chemical formula, but a different arrangement of the atoms, are **isomers**.



Lesson 2 – Chiral Centers

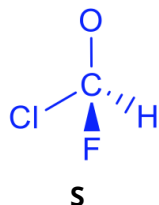
A **chirality center** is a carbon center that contains four unique substituents.

When using the **R,S naming system**:

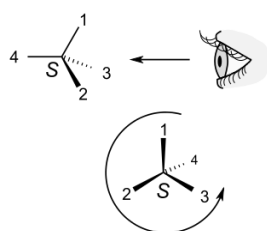
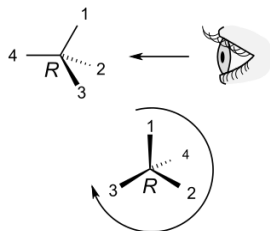
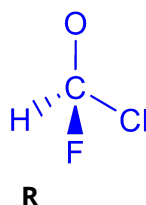
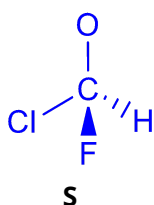
1. Find your stereocenter atom.
2. Prioritize the four appendages coming off the stereocenter atom using the **Cahn-Ingold-Prelog system**.
 - a. Highest priority = highest atomic number.
 - b. If there's a tie, keep going out in both directions, one by one, until the tie is broken.
3. Number your substituents 1, 2, 3, 4 (1 = highest priority, 4 = lowest priority)
4. Direct the lowest-priority substituent three-dimensionally away from you.
5. Make a circle from substituent 1 to 2 to 3.
 - a. Clockwise = R
 - b. Counterclockwise = S

Examples:

Chiral carbon



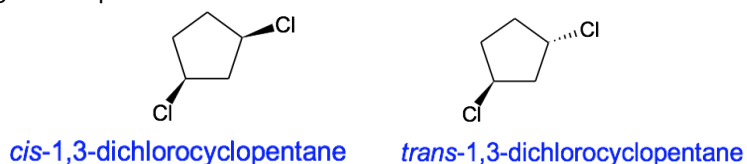
Enantiomers



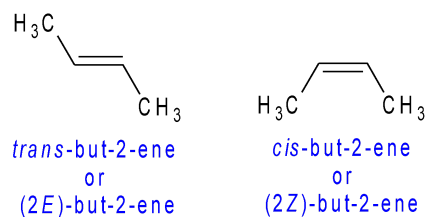
Lesson 3 - Diastereomers

There are three types of diastereomers:

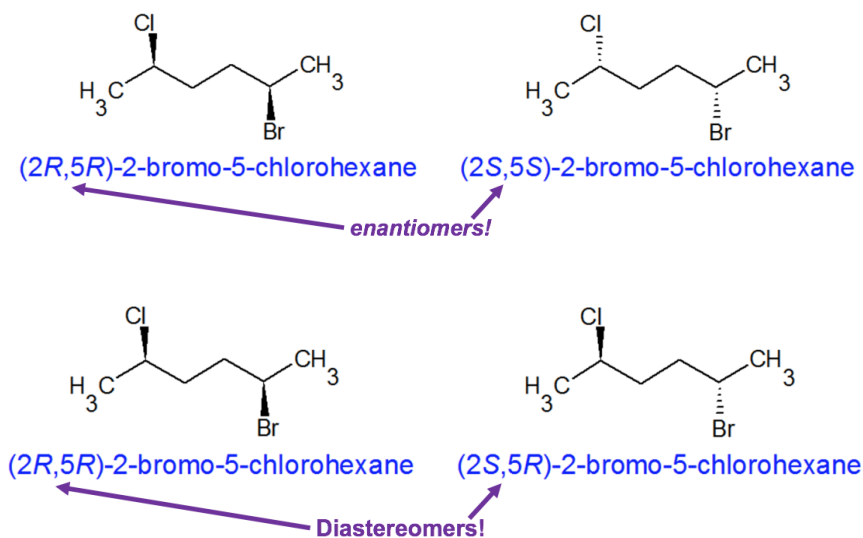
1. Cis/trans isomers of ringed compounds:



2. Cis/trans or E/Z isomers of alkenes:



3. Stereoisomers with multiple stereocenters that do NOT have exactly-opposite R,S configurations, and are NOT mirror images of one-another. If stereoisomers are mirror images of one-another (have exact opposite R,S assignments), they are considered **enantiomers**.



Lesson 4 – Counting Stereoisomers

When counting how many stereoisomers one chiral molecule can possibly have, use the equation:

$$\# \text{ of possible stereoisomers} = 2^n$$

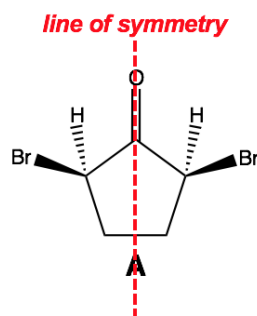
Where “n” is the number of chiral centers.

Lesson 5 – Chirality and Physical Properties

- Chiral molecules have the ability to rotate **plane-polarized light** when they're placed in a special machine called a **polarimeter**.
- Molecules that don't rotate plane-polarized light are called **achiral** or **inactive**. There are three kinds of optically-inactive (or **achiral**) molecules or situations:
 - If you have a 50/50 mixture of two enantiomers, then that mixture (called a **racemic mixture**) is **achiral**, despite all individual molecules being chiral!
 - Molecules that DON'T have stereochemistry in them (because they don't have **stereocenters**, or they don't have **cis/trans** stereochemistry in them) are **achiral**.
 - Meso** compounds are **achiral**.
- Enantiomers** share extremely similar physical properties, and may only be distinguished by the direction that they polarize light, and the way they interact with biological systems (some physiological enzymes may react with R and not S of a particular molecule).
- Diastereomers** have very different physical properties, and are separated easily, such as by boiling points.

Lesson 6 – Meso Compounds

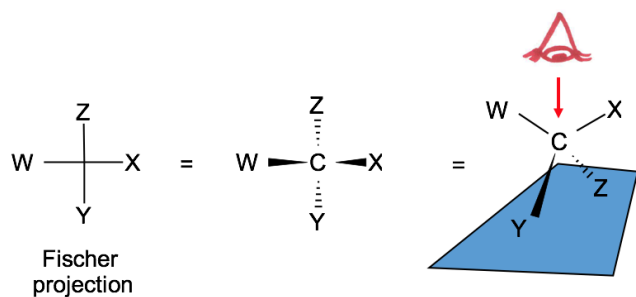
A **meso compound** is any molecule with two or more chirality centers, and a line of symmetry. An enantiomer of a meso compound is exactly the same as the original molecule (the two ARE superimposable).



Lesson 7 – Fischer Projections

Fischer projections are flat representations of three-dimensional molecules. They are especially useful for assessing chirality. Horizontal lines are used to represent atoms toward us, while vertical lines are used to represent atoms away from us.

Example:



Ch. 5: Spectroscopy

Lesson 1 – IR Spectroscopy

IR Spectroscopy Peaks You Should Memorize

If you see the following on your IR spectrum:	Then your compound has a:
Small to medium peak at 1600-1700 cm^{-1}	C=C or C=N
A BIG pointy peak at 1700 $\pm$ 50 cm^{-1}	C=O (carbonyl)
A LARGE, broad trough far to the left for alcohols and on top of 3000 cm^{-1} for carboxylic acids	OH
Big, pointy peaks coming straight down around 3000 cm^{-1}	C-H's
A sharp peak to the left of 3000 cm^{-1} (around 3200-3500 cm^{-1})	N-H (one peak for -NH, two peaks for -NH ₂)
Medium-sized peak at ~2200 cm^{-1}	C $\equiv$ N (a nitrile)
Vampire teeth at 1500-1600 cm^{-1} and 1300-1400 cm^{-1}	NO ₂

- IR spectroscopy is used most often to determine a molecule's functional groups.

Lesson 2 – UV Vis and Mass Spectrometry

- UV-Vis spectroscopy** is used mostly to analyze compounds with conjugated double bonds.
- Mass spectrometry** is a technique used to determine a compound's mass and isotopic patterns. For example, chlorine and bromine isotopes produce distinct M and M+2 peaks due to their natural abundance ratios:
 - Chlorine exhibits a 3:1 ratio due to the natural abundances of chlorine-35 and chlorine-37, respectively.
 - Bromine exhibits a 1:1 ratio due to the natural abundances of bromine-79 and bromine-81, respectively.

Lesson 3 – Degrees of Unsaturation

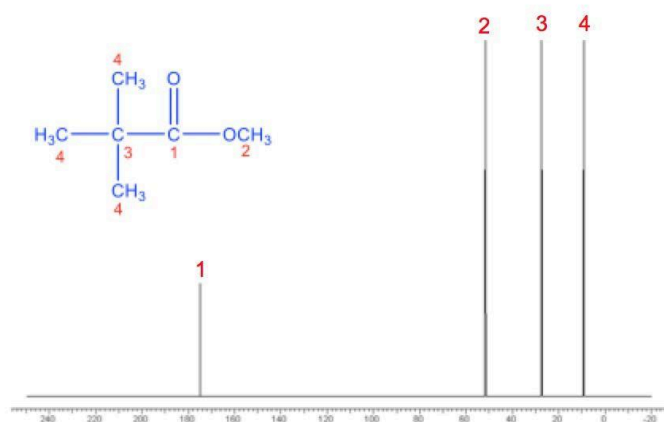
Degrees of unsaturation:

$$\# \text{ of degrees of unsaturation} = \# \text{ of double bonds or rings} = (\mathbf{A} - \mathbf{B})/2$$

Where **A** is the number of hydrogen atoms your compound *would* have if it didn't have any double bonds or rings ($\text{C}_n\text{H}_{2n+2}$), and **B** is the number of hydrogen atoms your compound in question *actually* has.

Lesson 4 – Degrees of Unsaturation Molecular Formula

$$\text{Degrees of unsaturation} = C - \left(\frac{H}{2}\right) - \left(\frac{X}{2}\right) + \left(\frac{N}{2}\right) + 1$$

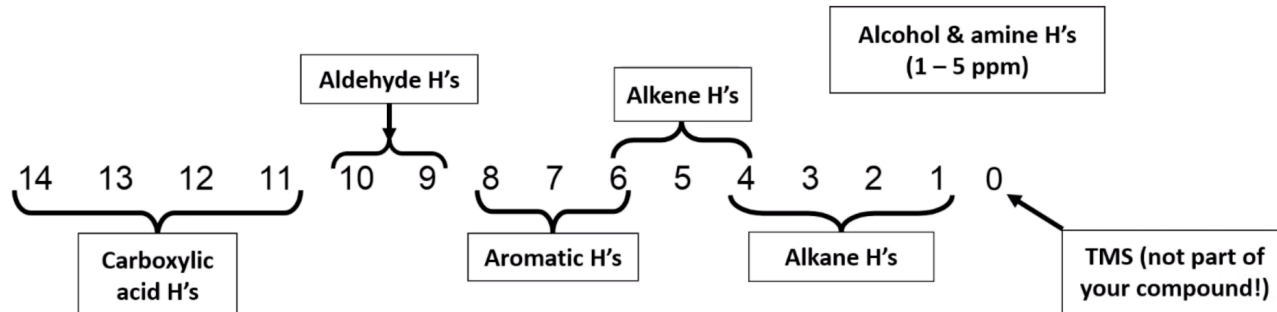
Lesson 5 – ^{13}C -NMR Spectroscopy

The labeled peaks are each produced by a different kind of carbon atom. By learning where different kinds of carbons show up on a ^{13}C -NMR spectrum, we can deduce compounds' structures.

- The more positively-charged an individual carbon is, the further to the left it will appear on an ^{13}C spectrum. The more negatively-charged, the further to the right it will appear.
- If you don't have a $\text{C}=\text{O}$ bond in your compound, you can basically ignore this.

Kind of carbon in C-NMR:	Where it shows up:
Carbonyl ($\text{C}=\text{O}$) carbons • Esters, amides, and carboxylic acids • Aldehydes and ketones	160-180 ppm >200 ppm
TMS (tetramethylsilane) – not part of your compound!	0 ppm

Lesson 6 – ^1H -NMR Spectroscopy



- Each “different kind of” (or “non-equivalent”) hydrogen gives a different signal. The more positively-charged, the further to the left it appears (**downfield**).
- The integral numbers above each peak tell you how many hydrogens are in that peak.
- Hydrogens get split by neighboring hydrogens. To figure out the splitting, count all the hydrogens next-door in all directions and add 1 (**$n+1$ rule**).

Ch. 6: IM Forces and Lab Techniques

Lesson 1 – Intermolecular Forces

Intra-molecular forces (forces **WITHIN** a molecule)

1. Covalent bonds: two non-metal atoms bond together and share electrons
2. Ionic bonds: metals bond to nonmetals and a transfer of electrons occurs
3. Metallic bonds: metal atoms bond together and electrons flow freely around their nuclei

Intermolecular forces (forces **BETWEEN** molecules)

1. Ion-dipole: ionic compounds interacting with polar compounds
2. Hydrogen bonding: H-O, H-N, or H-F
3. Dipole-dipole: Examples: H-Cl, C-O, S-H
4. Dispersion Forces: hydrocarbons, single elements, nonpolar molecules

Lesson 2 – Effect of IM Forces on Physical Properties

Type of Solid	Interaction	Properties	Examples
Metallic	 Metallic Bonding	Variable hardness and melting point, conductive	Fe, Mg
Ionic	 Ionic	High melting point, brittle, hard	NaCl, MgO
Network	 Covalent Network	High melting point, hard, non-conductive	C (diamond, graphite), SiO₂ (quartz)
Molecular	H-bonding, dipole-dipole, London dispersion	Low melting point, non-conductive	H₂, CO₂

The stronger a molecule's **intermolecular forces**:

- The higher its boiling point ↑
- The higher its melting point ↑
- The lower its vapor pressure ↓

Lesson 3 – Melting Points and Extractions

Melting point helps determine a compound's purity

Extraction techniques rely on using polar and nonpolar solvents

Lesson 4 – Acid-Base Extractions

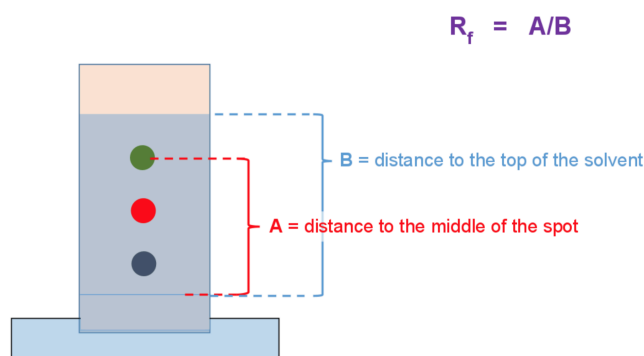
- For **carboxylic acids**, extract with aqueous **NaOH** or **NaHCO₃**
- For **phenols**, extract with aqueous **NaOH**
- For **amines**, extract with aqueous **HCl**

Lesson 5 – Distillation and Recrystallization

- **Distillation** separates mixtures of two or more volatile liquids
- **Fractional distillation** is used when the two volatile liquids have boiling points that are close together
- **Recrystallization** dissolves an impure compound in hot solvent and gradually precipitates the pure compound as the solution cools down.
 - o Note: If you're cooling down your product in recrystallization and no crystals form, you can scratch the side of the glass to provide a nucleation site to induce recrystallization. You can also use a **seed crystal**.

Lesson 6 – Chromatography

- **Gas-liquid chromatography** (or gas chromatography) is used to determine the relative abundance of each compound in a liquid mixture.
 - o Separates components in liquid mixture by boiling point.
 - o Lowest boiling point comes off the fastest.
- **Thin Layer Chromatography** (TLC) separates compounds by their solubility in the solvent (polarity). Most soluble compound travels the fastest and furthest up the plate.
 - o Usually uses a polar plate and nonpolar solvent.
 - o Compound that travels the furthest with nonpolar solvent is the most nonpolar compound.
 - o Retention Factor (R_f) is a number we use to tell how far up the TLC plate a compound travels.



- **Column Chromatography** (silica gel chromatography) separates compounds by polarity. The most nonpolar compounds travel the fastest, and the most polar compounds travel the slowest.
 - o Uses silica gel (polar) and eluent (impure mixture to be separated).
 - o The eluent is poured into the top of the column, and the most nonpolar compound travels down the fastest.
 - o Similar to Thin Layer Chromatography, except the most polar compound is at the top of the column, and the most nonpolar compound is at the bottom.

Lesson 7 – Qualitative Analysis

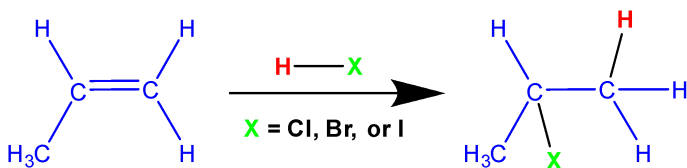
You also need to know commonly used OC Lab Tests for the DAT. Memorize the summary chart below.

Test	Reagent	Functional Group Tested	Positive Result
Tollens' test	$\text{Ag}_2\text{O} / \text{NH}_3$ or $\text{Ag}(\text{NH}_3)_2^+$	Aldehydes	Sides of flask are coated with a silver mirror
Iodoform test	I_2 / OH^-	Methyl ketones	Yellow precipitate forms (CHI_3)
Silver nitrate in alcohol	AgNO_3 in alcohol	Alkyl halides	Precipitate of Ag compound formed
Bromine test	$\text{Br}_2 / \text{CCl}_4$	Alkenes and alkynes	Brown color of bromine disappears
Jones test	$\text{CrO}_3 / \text{H}_2\text{SO}_4$	1° and 2° alcohols	Orange reagent turns blue-green

Ch. 7: Reactions of Alkenes and Alkynes

Lesson 1 – Alkene Additions and Hydrohalogenations

Hydrohalogenation (adding HX)



- The **H** goes on the carbon with more H's on it. The **X** goes on the carbon with fewer H's on it. This is called the **Markovnikov product**.

What's added: H^+ and Br^-

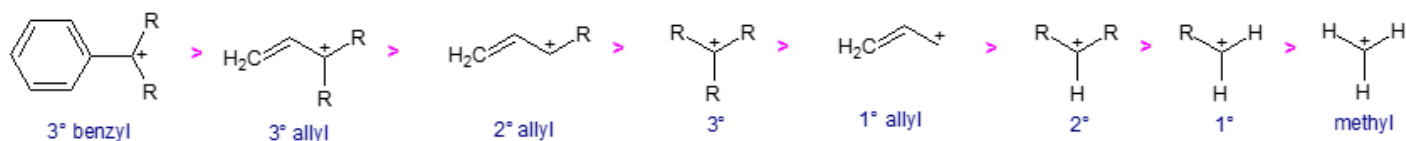
Regioselectivity: Markovnikov

Intermediate: carbocation

Rearrangements: possible

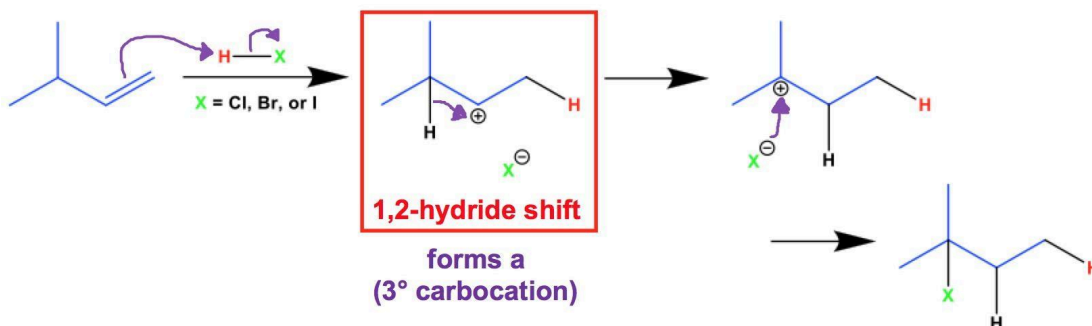
Carbocation stability

R = any alkyl
(hydrocarbon) chain

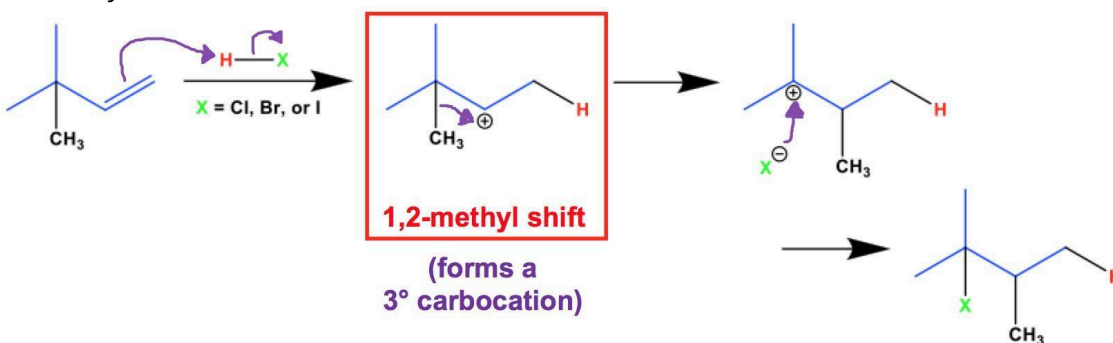


Lesson 2 - Carbocation Rearrangements

1,2-Hydride Shifts

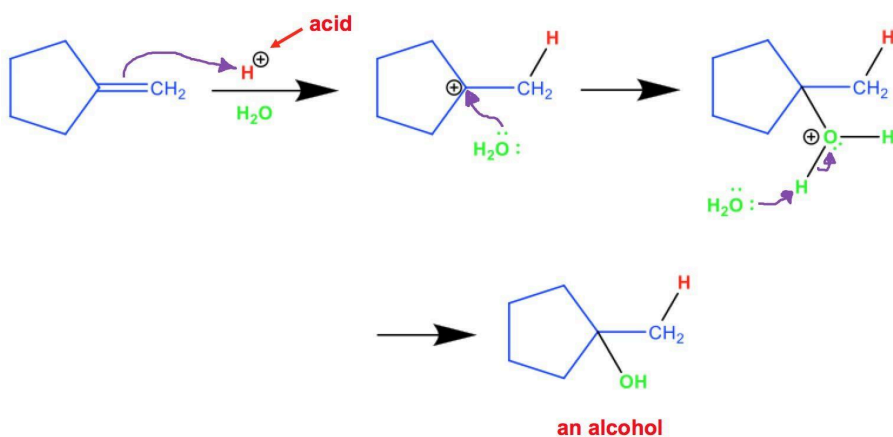


1,2-Methyl Shifts



Lesson 3 - How to add -OH and -OR to Alkenes

Acid-Catalyzed Hydration



What's added: H^+ and OH^-

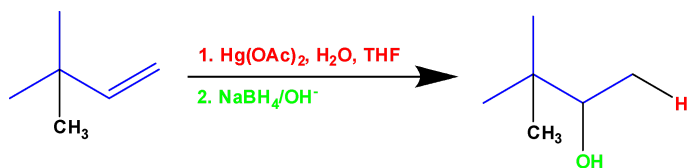
Regioselectivity: Markovnikov

Intermediate: carbocation

Rearrangement: possible

Lesson 3 - How to add -OH and -OR to Alkenes (Continued)

Oxymercuration-Demercuration



What's added: H^+ and OH^-

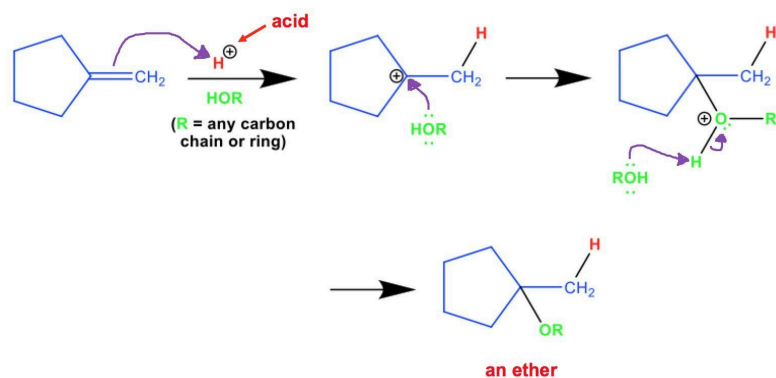
Regioselectivity: Markovnikov

Stereoselectivity: Anti addition

Intermediate: mercurinium ion

Rearrangement: **Not possible**

Acid-Catalyzed Alcohol Addition



What's added: H^+ and OR^-

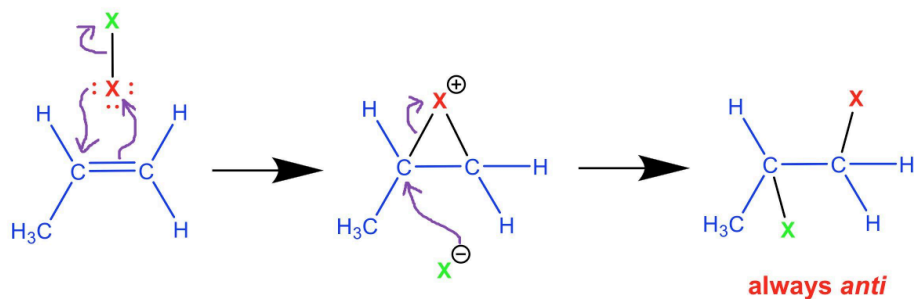
Regioselectivity: Markovnikov

Intermediate: carbocation

Rearrangement: possible

Lesson 4 - Adding Halogens to Alkenes

Adding Halogens



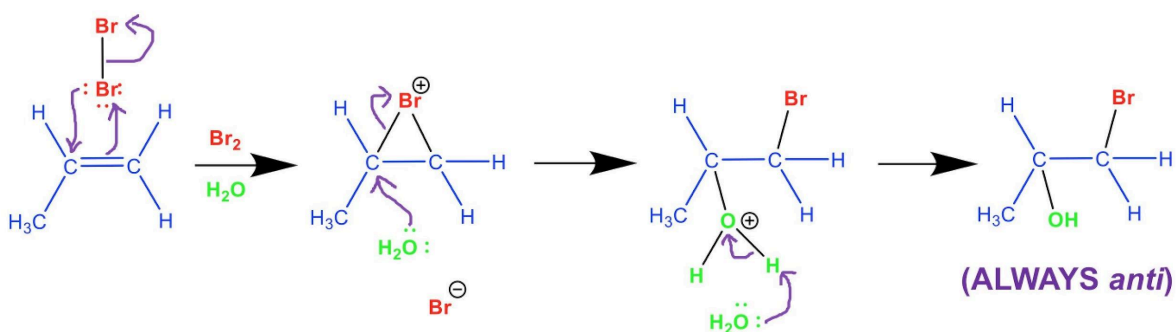
What's added: Br^+ and Br^- (or Cl^+ and Cl^-)

Stereoselectivity: anti

Intermediate: bromonium ion

Rearrangement: Not possible

Adding Halogens and H_2O (or ROH)



What's added: Br^+ and OH^- (or Br^+ and OR^-)

Regioselectivity: Markovnikov

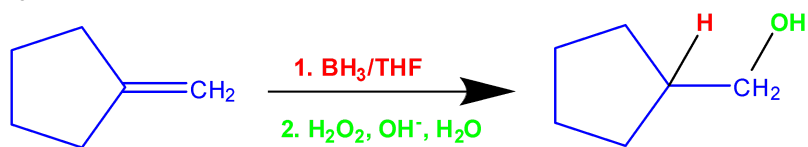
Stereoselectivity: Anti

Intermediate: bromonium ion

Rearrangement: Not possible

Lesson 5 - Anti-Markovnikov Alkene Additions

Hydroboration-Oxidation



What's added: H⁺ and OH⁻

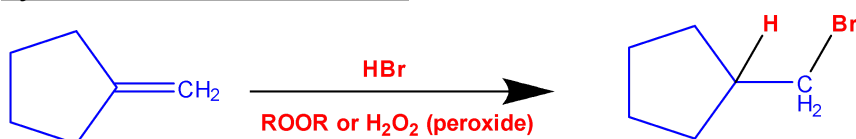
Regioselectivity: Anti-Markovnikov

Stereoselectivity: Syn

Intermediate: hydroxy-boranes

Rearrangement: Not possible

Hydrobromination with Peroxide



What's added: H[•] and Br[•]

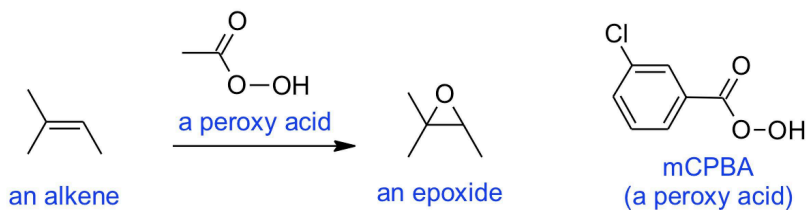
Regioselectivity: Anti-Markovnikov

Intermediate: radical

Rearrangement: Not possible

Lesson 6 - Epoxides and Dihydroxylations

Epoxide Reactions with Alkenes



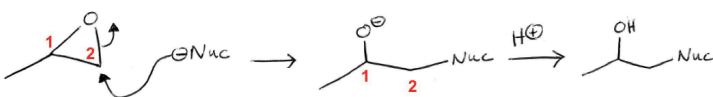
What's added: Oxygen

Stereoselectivity: syn

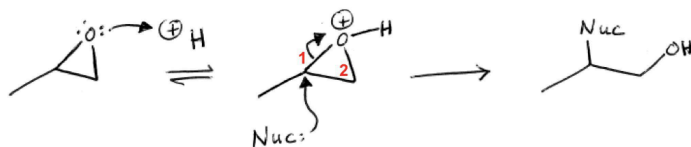
Rearrangement: Not possible

Anti-dihydroxylation

Under basic conditions:



Under acidic conditions:

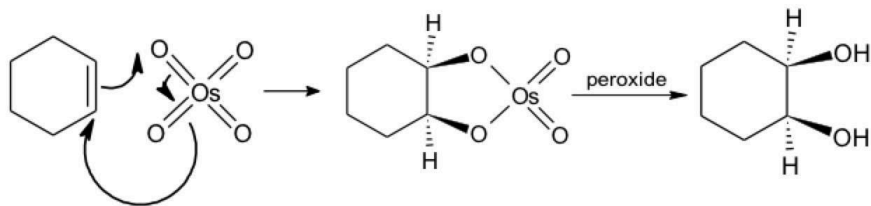


What's added: OH and OH

Stereoselectivity: *anti*

Rearrangement: not possible

Syn-dihydroxylation



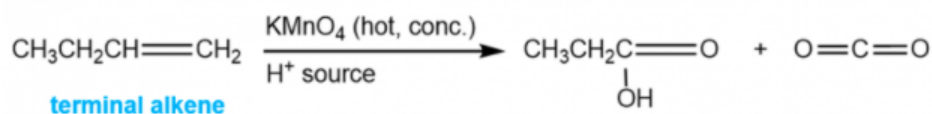
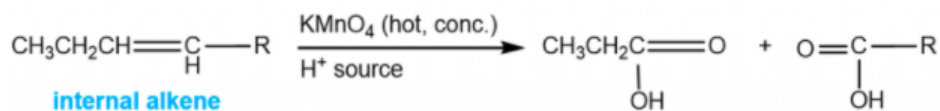
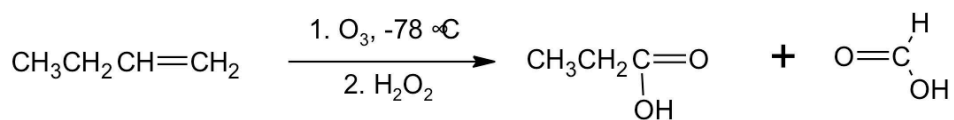
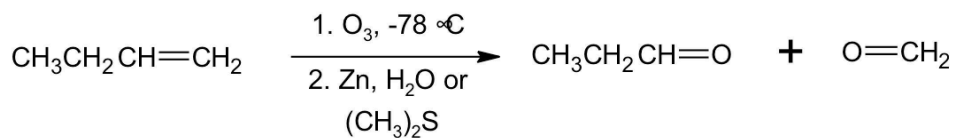
What's added: 2 OH's

Regioselectivity: 1,2

Stereoselectivity: Syn

Intermediate: osmate ester

Rearrangement: Not possible

Lesson 7 - OzonolysisOzonolysis

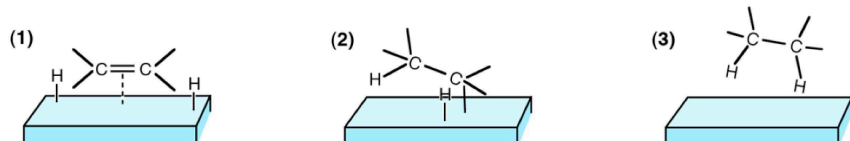
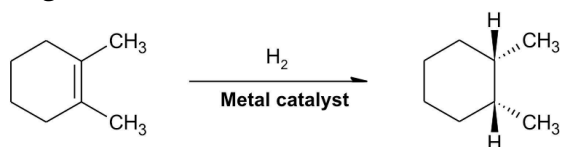
What happens: The C=C bond gets cut in half. An O gets placed on each half.

- If the workup (*step 2*) is Zn/H₂O or (CH₃)₂S, then that's it.
- If the workup (*step 2*) is H₂O₂, then one of the H atoms stuck to each alkene carbon gets replaced with an OH.
- Also, KMnO₄ (hot, conc.)/H₃O⁺ does the same thing as O₃ and H₂O₂.

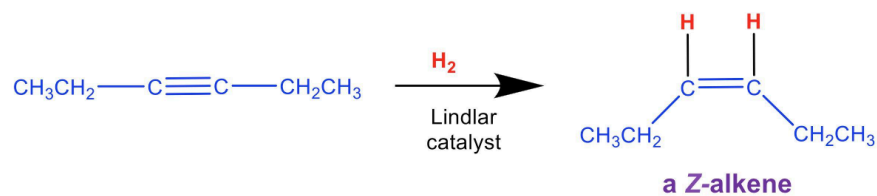
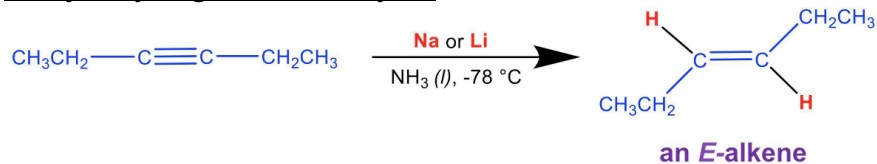
Stereoselectivity: *Syn*

Lesson 8 - Catalytic Hydrogenation

Catalytic Hydrogenation of Alkenes



Catalytic Hydrogenation of Alkynes



What's added: two H's

Stereoselectivity: Syn

Rearrangement: Not possible

Alkene + $\text{H}_2/\text{Pd}, \text{C} \rightarrow$ Alkane

Alkyne + $\text{H}_2/\text{Pd}, \text{C} \rightarrow$ Alkane

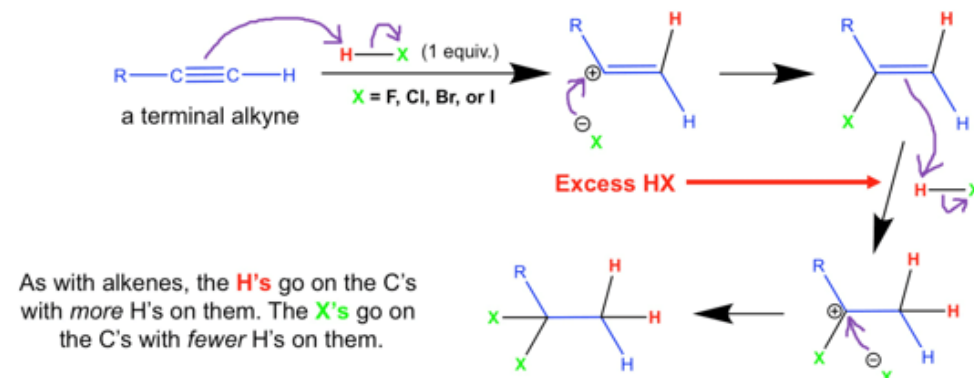
To stop at the *cis/trans* isomer of the alkene:

Alkyne + $\text{H}_2/\text{Lindlar's Catalyst} \rightarrow$ *cis* or *Z*-alkene

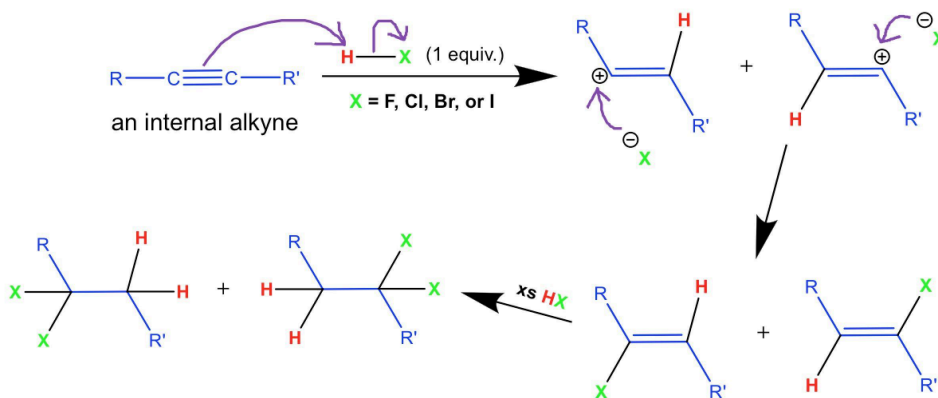
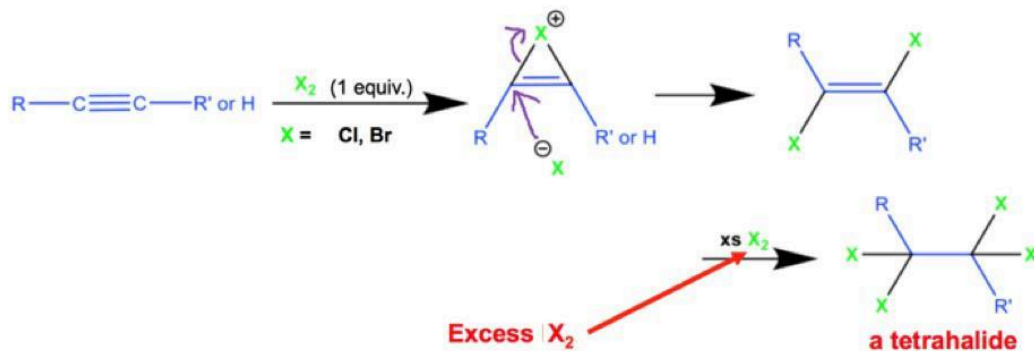
Alkyne + $\text{Na}/\text{NH}_3(l) \rightarrow$ *trans* or *E*-alkene

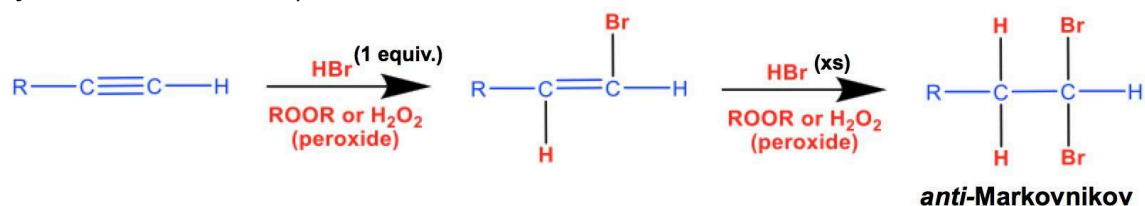
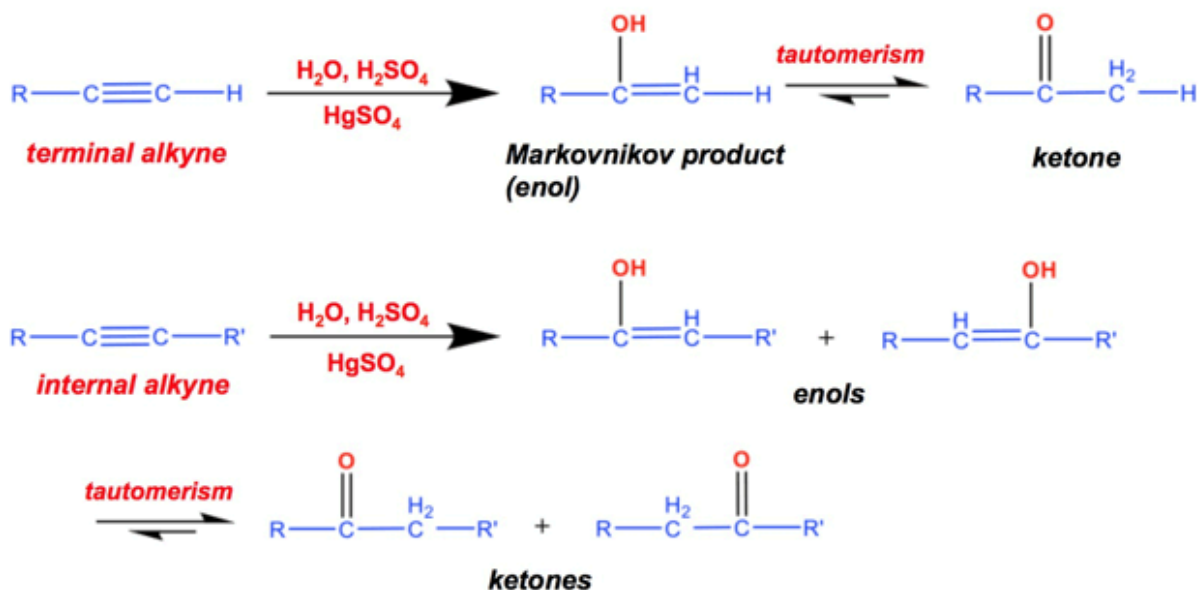
Lesson 9 – Alkyne Addition ReactionsHydrohalogenation of Alkynes

- Terminal alkynes

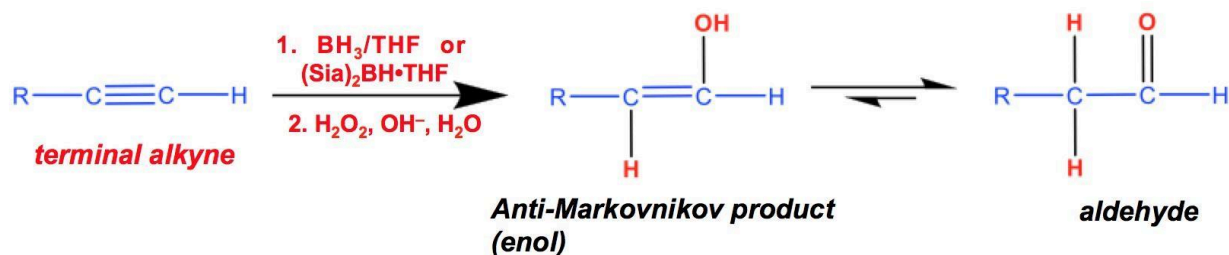
One Equivalent of HX:

- Internal alkynes (mixture of products)

One Equivalent of HX:Di-Halogenation of Alkynes**One Equivalent of X₂:**

Lesson 9 – Alkyne Addition Reactions (Continued)Hydrobromination with peroxideAcid-Catalyzed Hydration of Alkynes

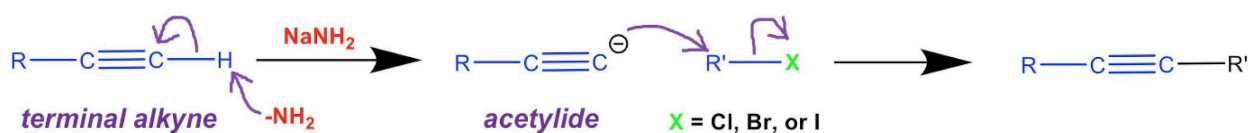
- **Reagent:** HgSO₄/H₂SO₄/H₂O
- You need a **Hg catalyst** for **terminal alkyne hydration**.
- This reaction adds an OH with Markovnikov regioselectivity to form an enol.
- The enol product then tautomerizes to form a **ketone**.

Lesson 10 – Alkyne Hydration and AlkylationAnti-Markovnikov Hydration of Alkynes

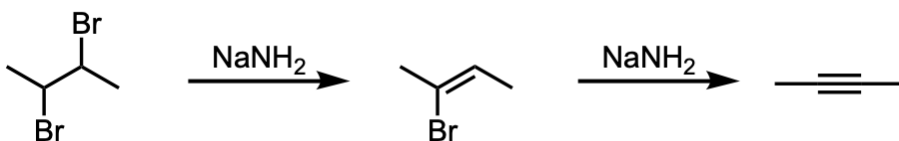
- **Reagents:** 1. $(\text{Sia})_2\text{BH}\cdot\text{THF}$ 2. $\text{H}_2\text{O}_2, \text{OH}^-, \text{H}_2\text{O}$
 - o (Note: you may also see $\text{BH}_3\cdot\text{THF}$, B_2H_6 , or even $(\text{Sia})_2\text{BH}$ drawn out. These all mean the same thing.)
- **Regioselectivity:** Adds OH Anti-Markovnikov to form an enol. This then tautomerizes to form an aldehyde.

Alkyne Hydration Summary

- Markovnikov conditions:
 - o Reagent: $\text{HgSO}_4/\text{H}_2\text{SO}_4/\text{H}_2\text{O}$
 - o You need a **Hg catalyst** for terminal alkyne hydration.
 - o This reaction adds an OH with Markovnikov regioselectivity to form an enol.
 - o The enol product then **tautomerizes** to form a **ketone**.
- Anti-Markovnikov conditions:
 - o Reagents: 1. $(\text{Sia})_2\text{BH}\cdot\text{THF}$ 2. $\text{H}_2\text{O}_2, \text{OH}^-, \text{H}_2\text{O}$
 - o (Note: you may also see $\text{BH}_3\cdot\text{THF}$, B_2H_6 , or even $(\text{Sia})_2\text{BH}$ drawn out. These all mean the same thing.)
 - o **Regioselectivity:** Adds OH Anti-Markovnikov to form an enol. This then **tautomerizes** to form an **aldehyde**.

Alkylation of Alkynes**Lesson 11 – Forming Alkynes from Alkenes**

NaNH_2 can remove H from a halogen substituted alkene, creating an alkyne.



Lesson 12 – Thermodynamic vs. Kinetic Products

Kinetic product:

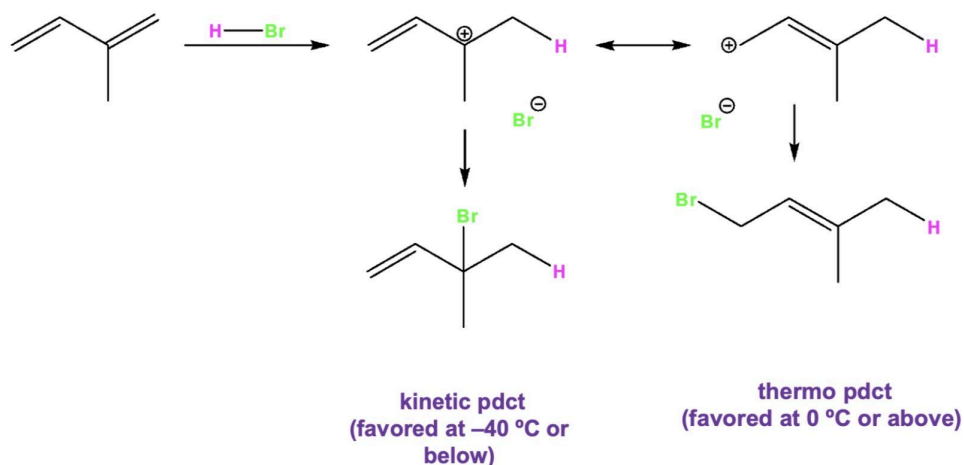
- Forms fastest
- Comes from the most stable carbocation intermediate
- Favored at low temperature (called *kinetic control*): $-40\text{ }^{\circ}\text{C}$ or lower

In a **conjugated diene** addition reaction, the **kinetic** product is the product that comes from the **most stable carbocation intermediate**.

Thermodynamic ("thermo") product:

- Forms more slowly
- Favored at higher temperature (called *thermodynamic control*): $\sim 0\text{ }^{\circ}\text{C}$ or higher
- Is the *most* substituted (most internalized) alkene product, which is the most stable final product, due to *Zaitsev's rule*.

In a conjugated diene addition reaction, the **thermo** product is the product that has the **most substituted double bond at the end**.



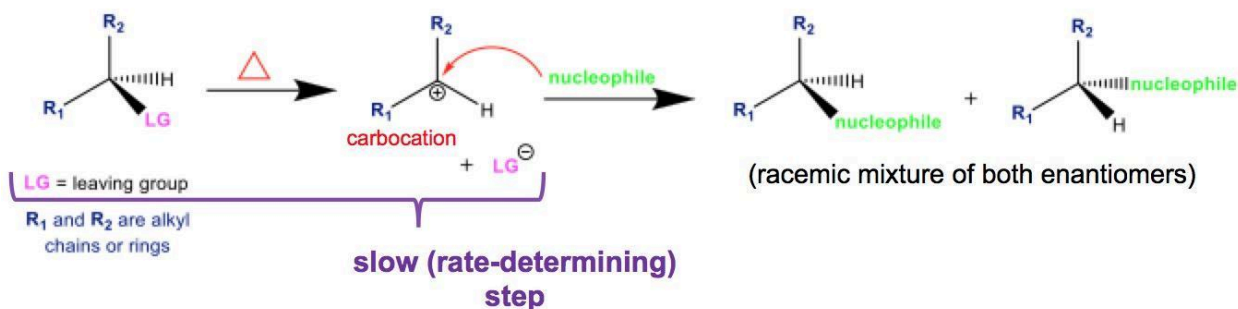
Ch. 8: Substitution and Elimination Reactions

Lesson 1 and 2 – Substitution Reactions

S_N1 Reaction – Substitution Nucleophilic Unimolecular

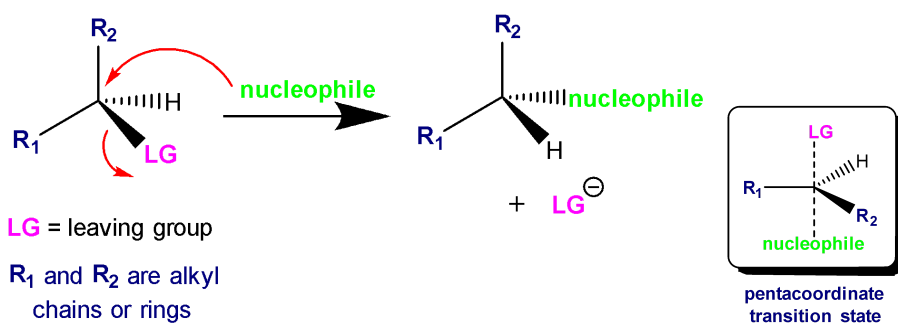
Rate = $k[\text{electrophile}]$

The **S_N1** reaction:



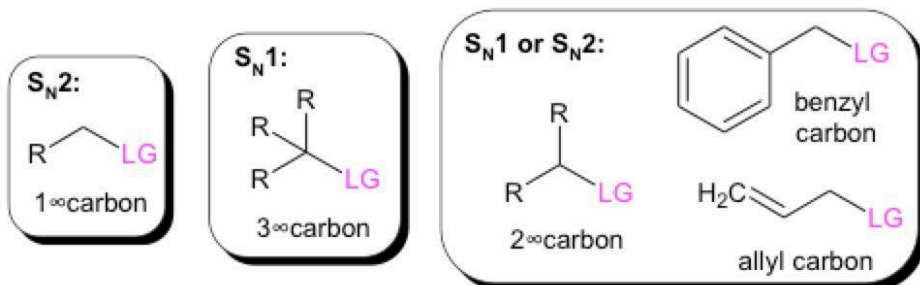
S_N2 Reaction – Substitution Nucleophilic Bimolecular

Rate = $k[\text{electrophile}][\text{nucleophile}]$



Lesson 1 and 2 – Substitution Reactions Basics (Continued)Choosing between S_N1 and S_N2 reactions

- Is the carbon bonded to the leaving group 1°, 2°, or 3°?



- Is my nucleophile strong or weak?
 - Strong** nucleophiles have localized negative charges.
 - Exceptions: Negative charges that are **resonance-stabilized** are weak.
 - Some **strong** nucleophiles (**S_N2 reactions**): CN⁻, OR⁻, OH⁻, RS⁻, NR₂⁻, R⁻
 - Some **weak** nucleophiles (**S_N1 reactions**): RCO₂⁻, HOR, H₂O, HSR, HNR₂

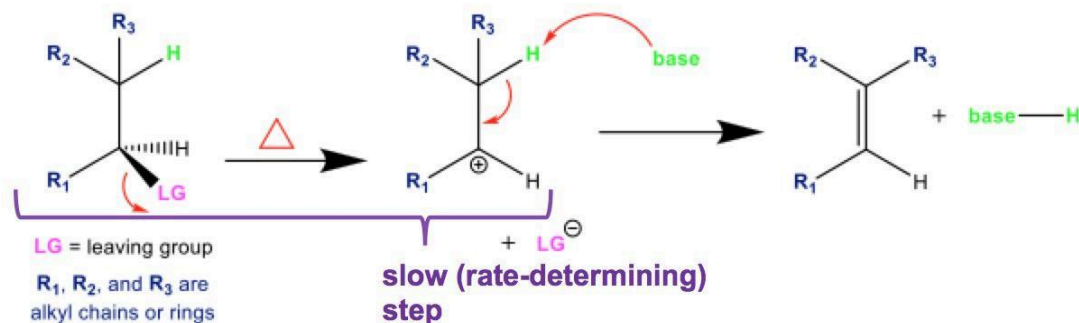
S_N2 vs. S_N1 reactions summary

	SN2	SN1
Nucleophile	strong	weak
Electrophile	Methyl > 1° > 2° (no 3°)	3° > 2° (no 1° or methyl)
Solvent	aprotic preferred	protic
Leaving Group	I > Br > Cl > F	I > Br > Cl > F
Rearrangement	Not possible	Possible (hydride/methyl shifts)
Inversion	Yes	No (racemization)

Lesson 3 and 4 – Elimination Reactions

E1 Reactions – Elimination Unimolecular

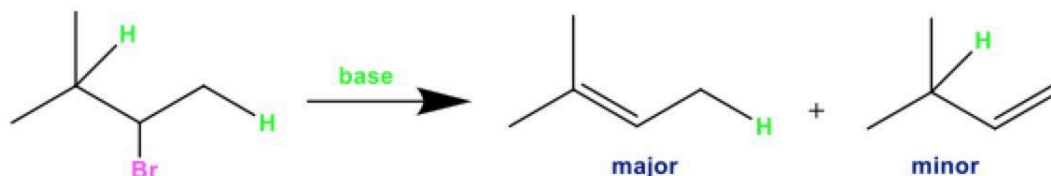
Rate = $k[\text{substrate}]$



- Forms most substituted double bond (Zaitsev's Rule)

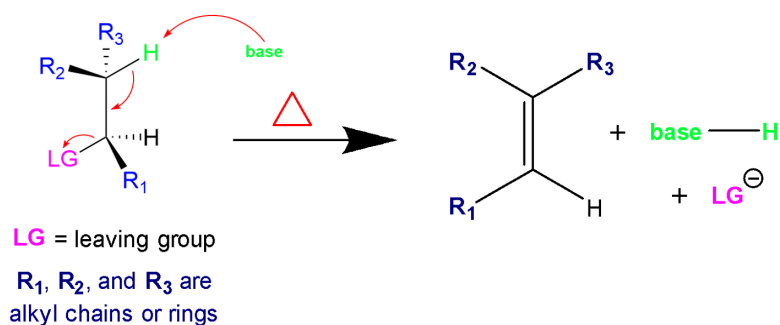
Zaitsev's Rule

- E1** and **E2** reactions give the **more substituted** C=C bond and favor the *E*-alkene.



E2 Reactions – Elimination Bimolecular

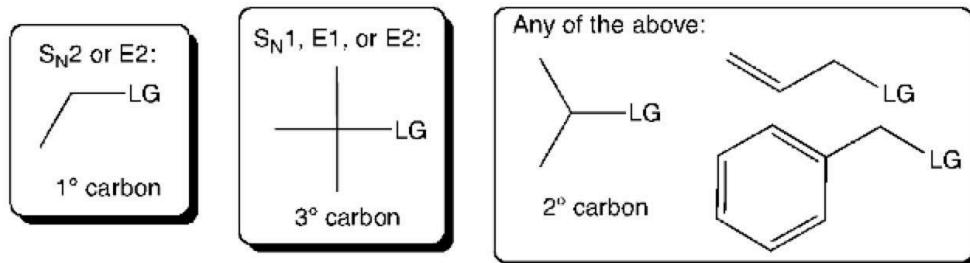
Rate = $k[\text{substrate}][\text{base}]$



- H and Leaving Group must be antiperiplanar (anti-coplanar)
- Forms most substituted double bond (Zaitsev's Rule)
- Forms least substituted double bond if bulky base is used like $t\text{-butoxide } (\text{CH}_3)_3\text{CO}^-$

Lesson 3 and 4 – Elimination Reactions (Continued)Choosing between E1 and E2 reactions

1. Is the carbon bonded to the leaving group 1°, 2°, or 3°, or stabilized?



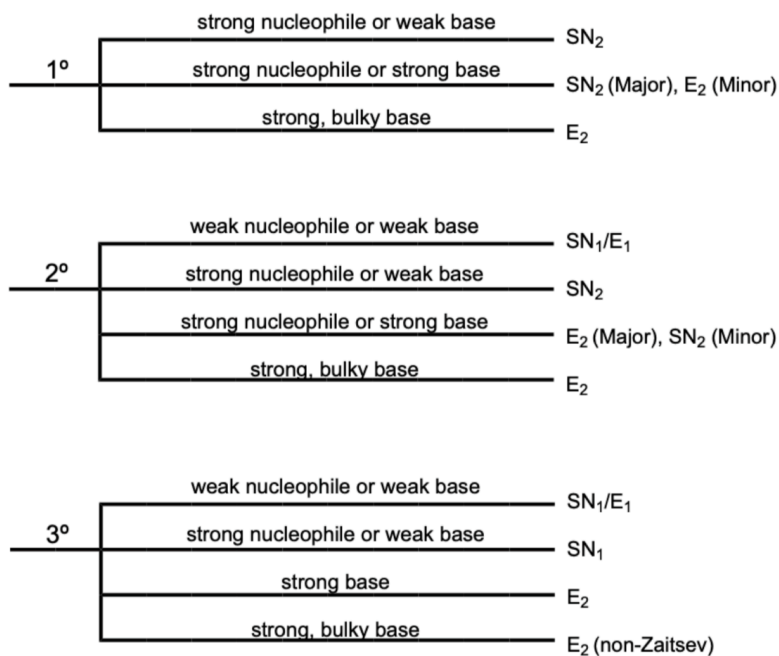
2. Is my base strong or weak?
- Strong** bases have negative charges.
 - Exceptions: Negative charges that are **resonance-stabilized** are weak.
 - Some **strong** bases (**E2 reactions**): OR^- , OH^- , NR_2^-
 - Some **weak** bases (**E1 reactions**): RCO_2^- , HOR , H_2O , HNR_2

E2 vs. E1 reactions summary

	E2	E1
Base	strong	weak
Electrophile	$3^\circ > 2^\circ > 1^\circ$	$3^\circ > 2^\circ$ (no 1° or methyl)
Solvent	aprotic preferred	protic
Leaving Group	$\text{I} > \text{Br} > \text{Cl} > \text{F}$	$\text{I} > \text{Br} > \text{Cl} > \text{F}$
Rearrangement	Not possible	Possible (hydride/methyl shifts)
Stereochemistry	Antiperiplanar	None

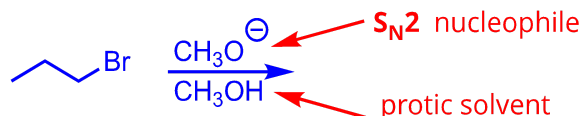
Lesson 5 – Choosing Between SN1, SN2, E1, E2Protic solvents = S_N1, E1, and E2 reactionsAprotic solvents = S_N2 and E2 reactionsProtic solvent examples: Anything with **hydrogen bonding**. Alcohols (methanol, ethanol), water, acetic acidAprotic solvent examples: DMSO (dimethylsulfoxide), DMF (dimethylformamide), THF (tetrahydrofuran), ether, acetone

	SN2	E2	SN1	E1
Electrophile	CH ₃ > 1° > 2°	3° > 2° > 1°	3° > 2°	3° > 2°
Nucleophile/ Base	Strong nuc	Strong base	weak nuc	weak base
Solvent	aprotic preferred	aprotic preferred	protic	protic

Substitution vs. Elimination FlowchartStrong Nucleophiles: CN⁻, OR⁻, OH⁻, RS⁻, NR₂⁻, R⁻Strong Bases: OR⁻, OH⁻, RS⁻, NR₂⁻, R⁻, (CH₃)₃CO⁻Weak Nucleophiles: RCO₂⁻, HOR, H₂O, HSR, HNR₂Weak Bases: RCO₂⁻, HOR, H₂O, HSR, HNR₂

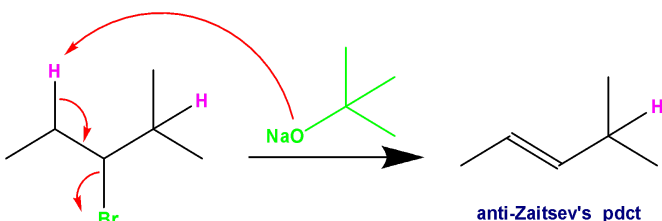
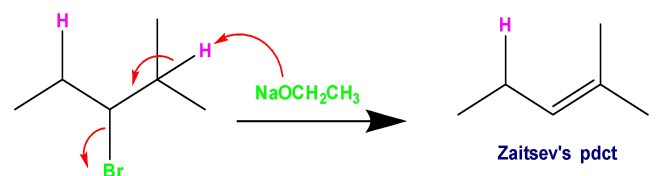
Lesson 6 – Exception to Substitution and Elimination**One Solvent Exception**

One exception to the **protic $\neq$ S_N2** rule is alkoxide bases being paired with their conjugate acids:



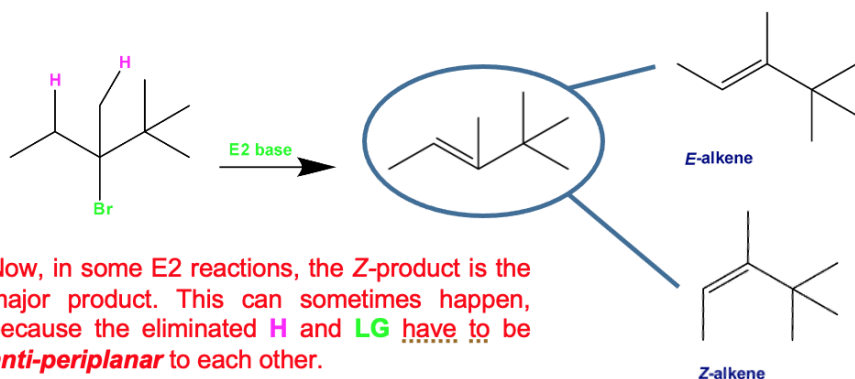
As it turns out, when using alkoxide nucleophiles, we often choose their protonated counterparts as our solvents.

When a strong S_N2 nucleophile like CH_3O^- is paired with its conjugate acid (CH_3OH) as solvent, the reaction will still go S_N2 even though the solvent is protic.

Exception to Zaitsev's Rule

When sterically-bulky tert-butoxy bases are used, **the less-substituted C=C bond (the anti-Zaitsev or Hofmann product) is favored.**

Sometimes the Z-product is formed because the H and LG have to be antiperiplanar to each other.

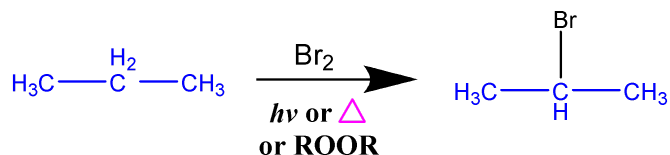


In **E2** reactions, the eliminated **H** must be in the same three-dimensional plane, but pointing 180° in the opposite direction, as the **leaving group**.

Ch. 9: Free Radical Halogenation and Diels Alder

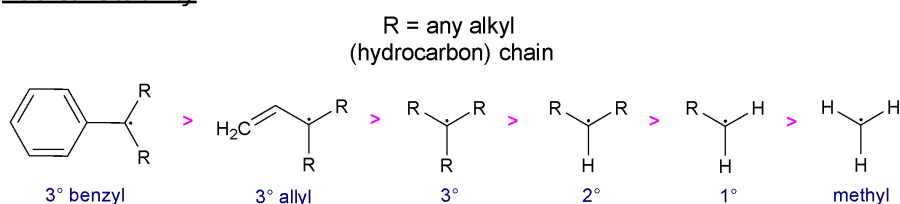
Lesson 1 – Free Radical Halogenation

Radical Reaction

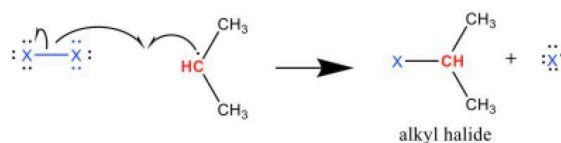
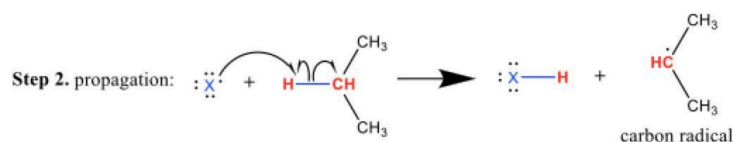
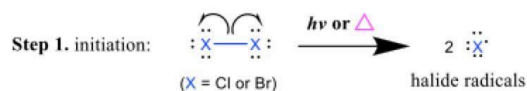


- **Bromine** is very selective, it likes the most stable radical.
- Chlorine isn't selective, that's why we usually use bromine in radical halogenation.

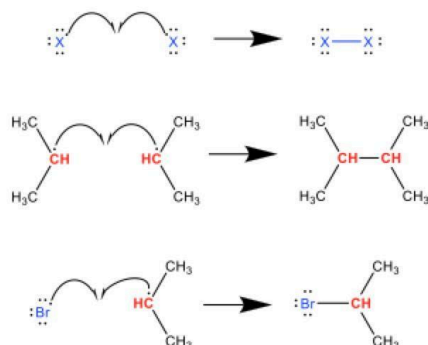
Radical Stability



Mechanism



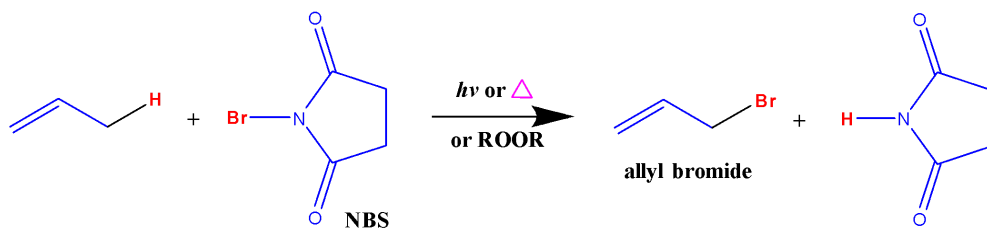
Step 3. termination (three forms):



- **Initiation** has no radicals on the left side, and radicals on the right side of the equation.
- **Propagation** has radicals on both sides of the equation.
- **Termination** has radicals on the left side, and no radicals on the right side of the equation.

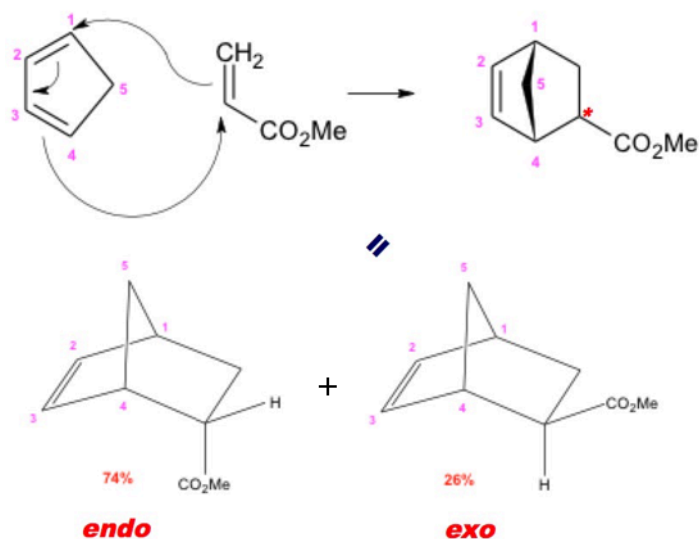
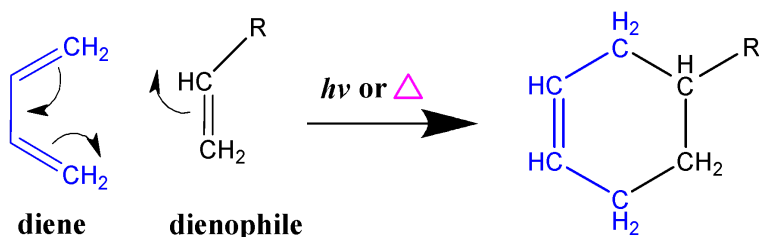
Lesson 2 – NBS in Radical Halogenation

- **NBS** (*N*-BromoSuccinimide) adds Br to the carbon that is one position away from the double bond (the allylic carbon).



Lesson 3 – Diels-Alder Reaction

- Concerted mechanism
- Pericyclic reaction



Ch. 10: Aromatic Compounds

Lesson 1 – How to Determine Aromaticity

1. It must be **cyclic** or **polycyclic**
 2. No sp^3 -hybridized atoms in the ring. The ring must be planar.
 3. The number of π electrons in the ring must equal **$4n + 2$** ($n = 0, 1, 2$, etc.). This is called **Hückel's rule**. Usually 2, 6, 10, 14 π electrons in the system.
- **Non-aromatic** compounds don't satisfy rules 1 or 2
 - **Anti-aromatic** compounds satisfy rules 1 and 2, but not rule 3

Lesson 2 – Effects of Aromaticity

- **Effects on S_N1** – If a molecule forms an aromatic carbocation intermediate during S_N1 , then it will react through S_N1 faster since the intermediate step is more stable.
- **Effects on Acidity** – If the conjugate base is more stable due to aromaticity, then the acid will be more acidic.

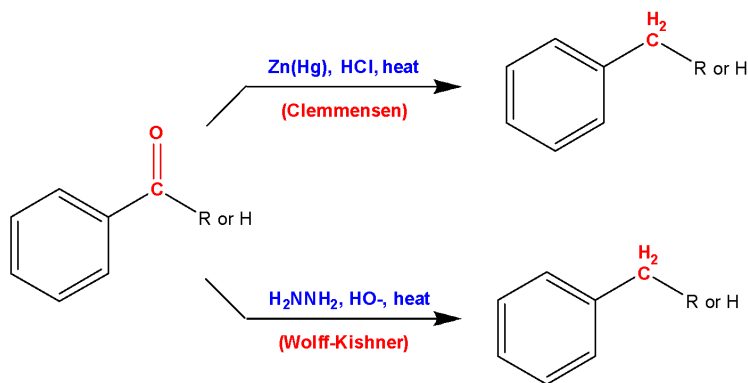
Lesson 3 – Side Reactions of Benzenes

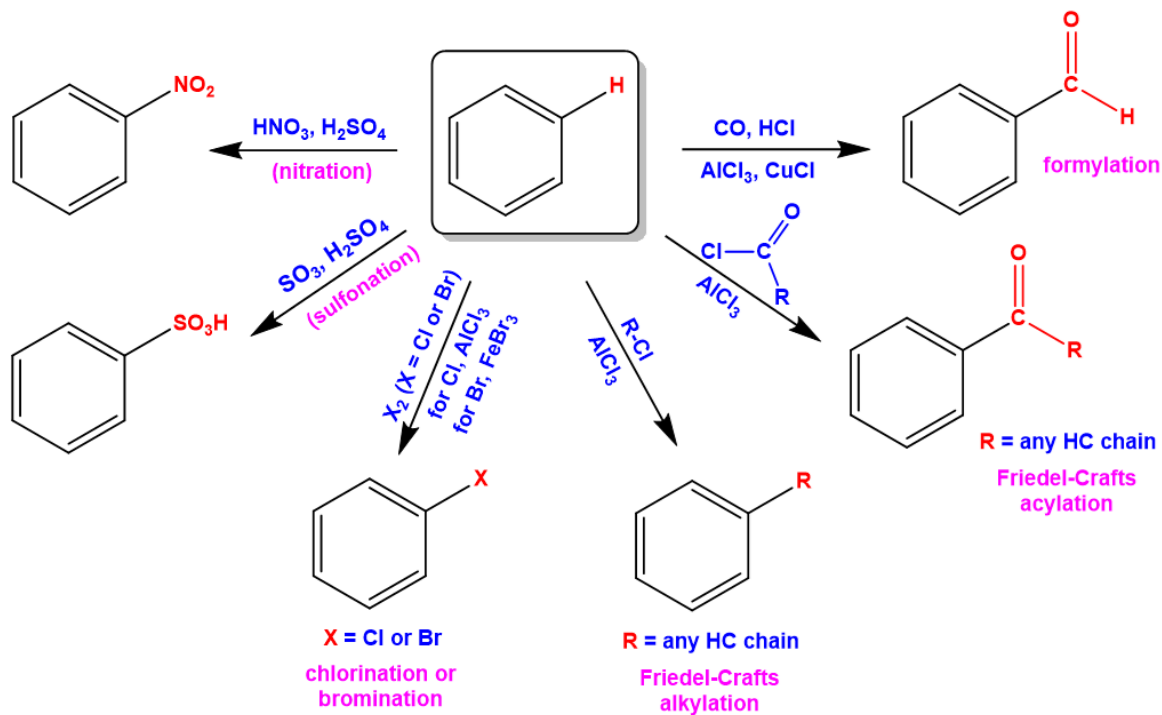
Side Chain Oxidation

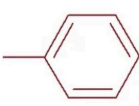


- You can only do this if your benzylic carbon is bonded to at **least one H**.

Side Chain Reduction

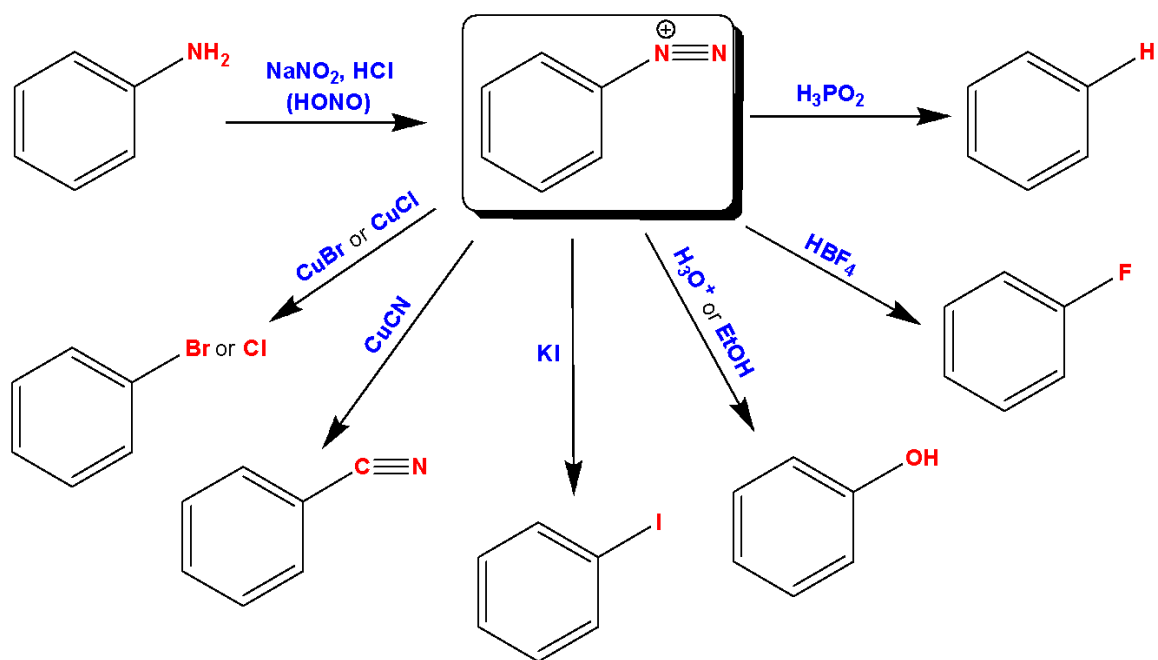


Lesson 4 – Electrophilic Aromatic Substitution (EAS)EAS ReactionsOrtho/Para and Meta Directors

Electron withdrawers		Electron donors (ortho/para directors)
meta directors	weak o/p directors	
$-\text{NO}_2$ $-\text{SO}_3\text{H}$ $-\text{C}\equiv\text{N}$ $-\text{C}(=\text{O})\text{CH}_3$ $-\text{C}(=\text{O})\text{OH}$ $-\text{C}(=\text{O})\text{H}$ $-\text{C}(=\text{O})\text{OR}$ $-\text{NR}_3^+$ R = alkyl or H	$-\ddot{\text{F}}:$ $-\ddot{\text{Cl}}:$ $-\ddot{\text{Br}}:$ $-\ddot{\text{I}}:$	$-\ddot{\text{N}}\text{H}_2$ $-\ddot{\text{O}}\text{H}$ $-\ddot{\text{O}}\text{R}$ $-\text{NH}-\text{C}(=\text{O})\text{R}$ $-\text{R}$  R = alkyl

Note: Halogens are a bit of an exception. Halogens are ortho/para directing, however because of their strong electronegativity, they are electron withdrawers.

Lesson 5 - Diazonium Salts

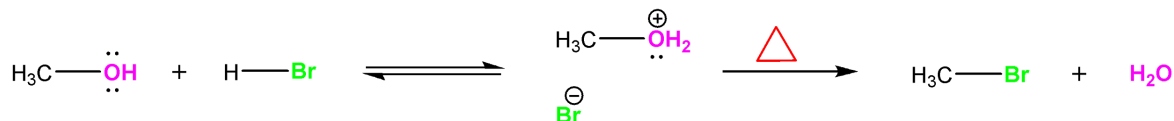


Ch. 11: Alcohols, Ethers, Epoxides

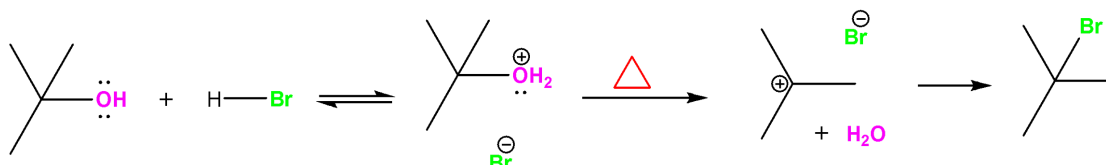
Lesson 1 – Substitution and Elimination of Alcohols

Substitution Reaction with H-X

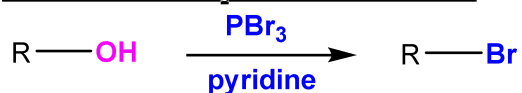
- Proceeds via S_N2 for 1° alcohols and methanol



- Proceeds via S_N1 for 3° and 2° alcohols

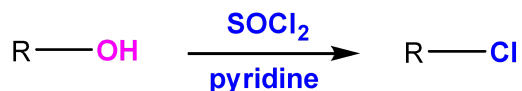


Reaction with PBr_3 (1° and 2° alcohols)



$\text{R} = \text{CH}_3, 1^\circ, \text{ or } 2^\circ \text{ (not } 3^\circ\text{)}$

Reaction with SOCl_2 (1° and 2° alcohols)

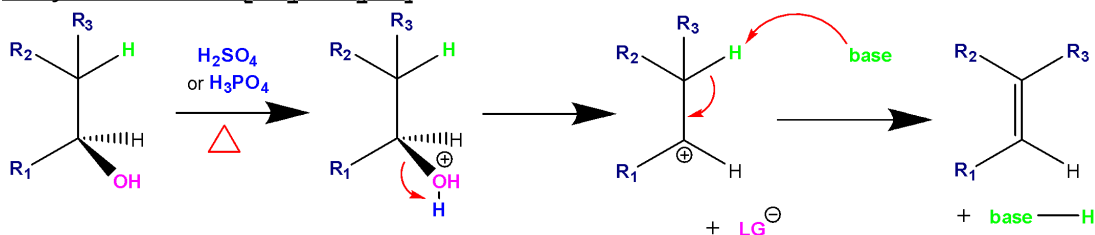


$\text{R} = \text{CH}_3, 1^\circ, \text{ or } 2^\circ \text{ (not } 3^\circ\text{)}$

Conversion to Sulfonate Esters (tosylates and mesylates)



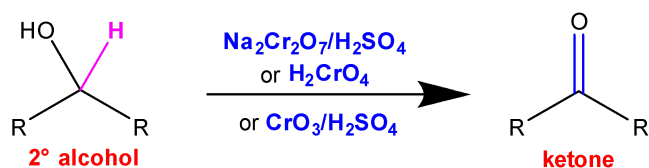
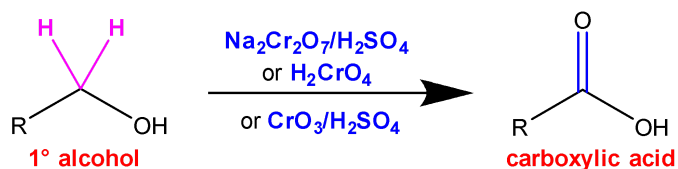
Dehydration with H_2SO_4 or H_3PO_4



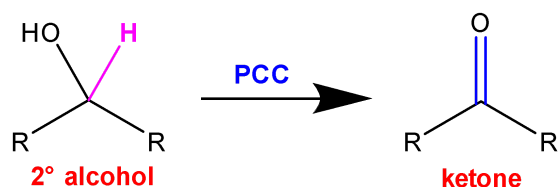
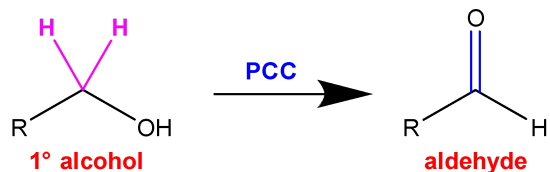
- E2** for 1° alcohols
- E1** for 3° and 2° alcohols

Lesson 2 - Oxidizing Alcohols

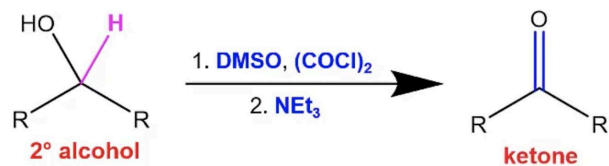
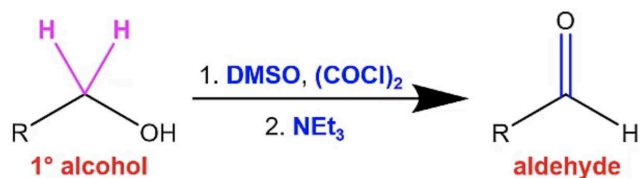
- **Chromium reagents** oxidize 1° alcohols and aldehydes to carboxylic acids
- 2° alcohols are oxidized into ketones

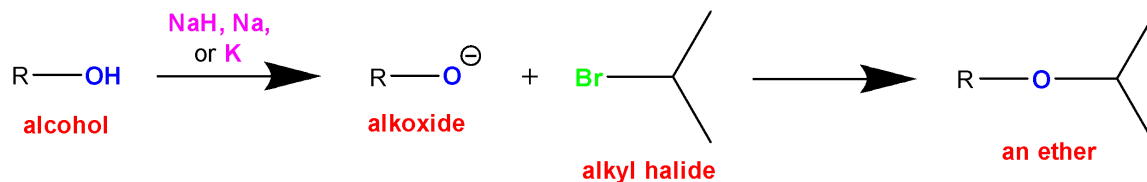


- **PCC** oxidizes 1° alcohols to aldehydes and 2° alcohols to ketones

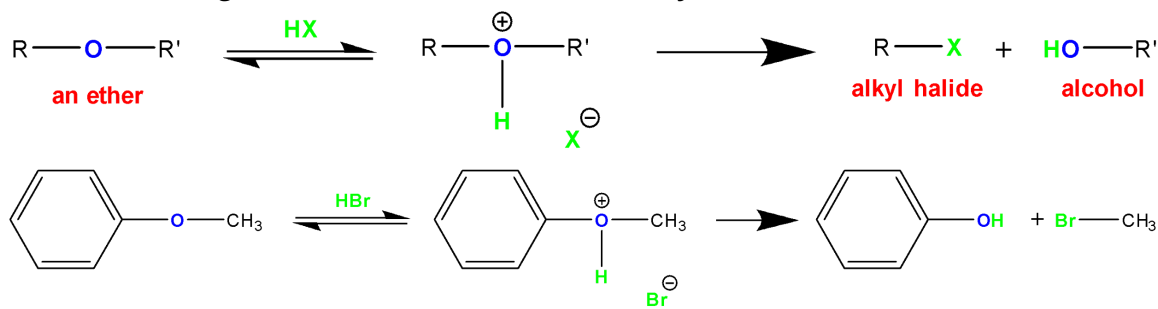
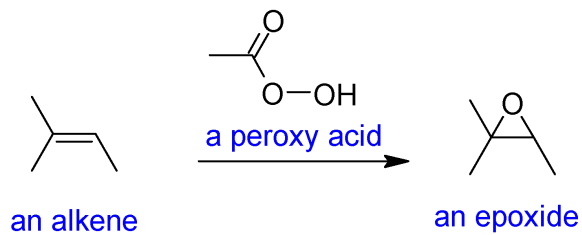
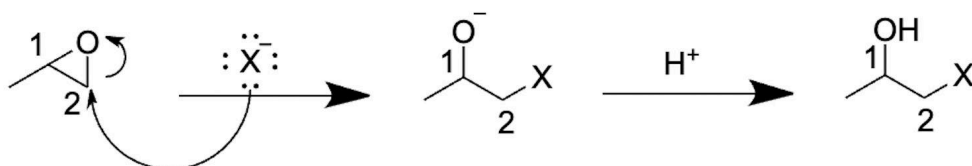
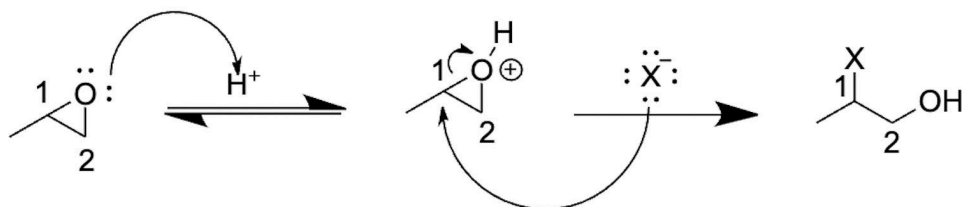


- **Swern oxidation** does the same thing as PCC, it oxidizes 1° alcohols to aldehydes and 2° alcohols to ketones



Lesson 3 - Reactions of EthersWilliam Ether Synthesis (S_N2)Adding H-X to Ethers

- The reaction will proceed through an **S_N1 mechanism** if you have a **secondary or tertiary carbon group**.
- If not, it'll go **S_N2** and attack the **least sterically hindered side**.

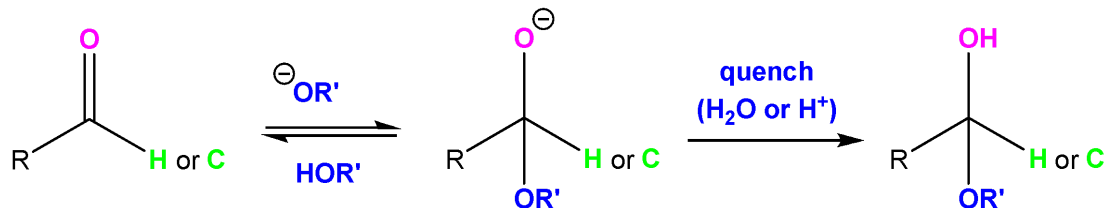
**Lesson 4 - Reactions of Epoxides**Epoxide SynthesisRing Opening of Epoxides**Base-Catalyzed** (Nucleophile attacks **less** substituted side)**Acid-Catalyzed** (Nucleophile attacks the **more** substituted side)

Ch. 12: Aldehydes and Ketones

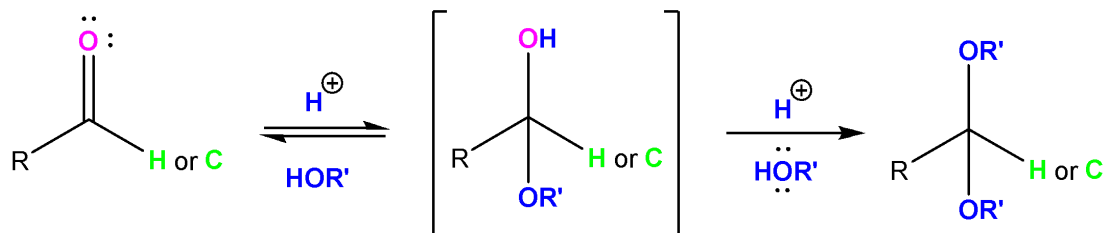
Lesson 1 – Alcohol Reactions with Carbonyls

Addition of Alcohols

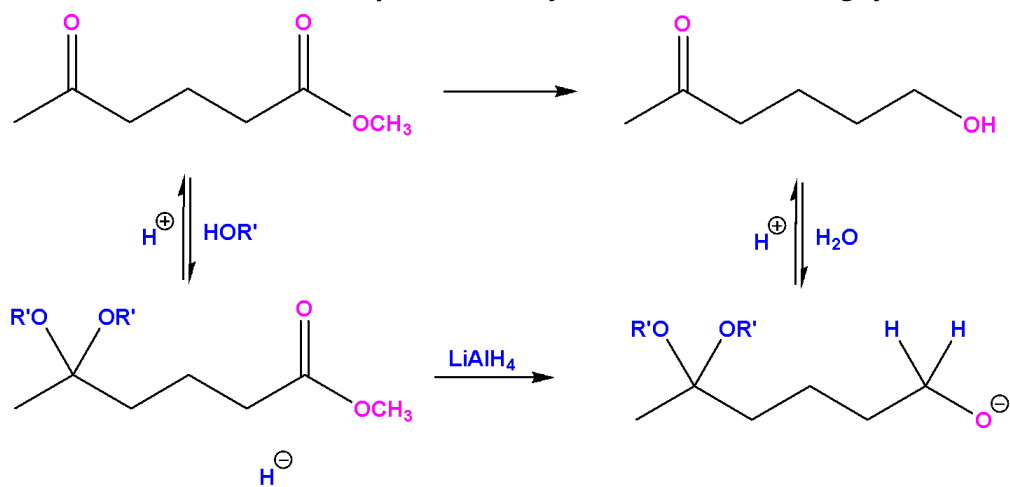
Base-catalyzed (forms hemi-acetal/hemi-ketals)



Acid-catalyzed (forms acetal/ketal)

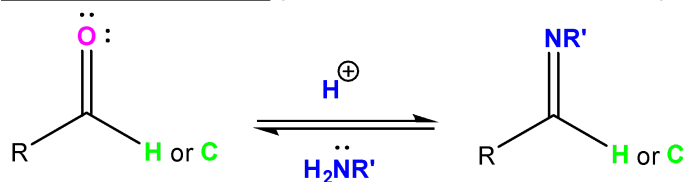


- Ketals** can be used to **protect** aldehydes and ketones during synthesis

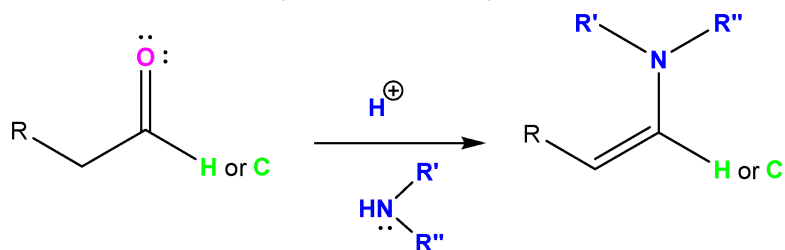


Lesson 2 - Amine Reactions with Carbonyls

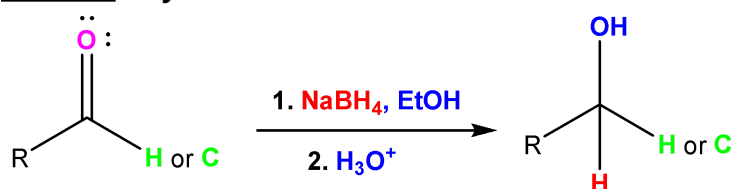
Addition of 1° Amines (forms Imines, aka Schiff bases)



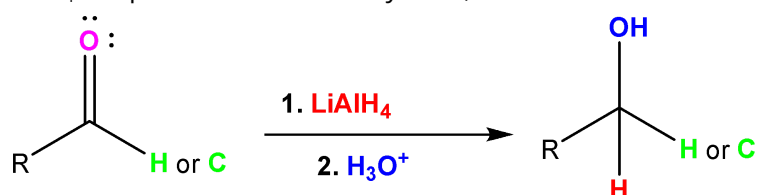
Addition of 2° Amines (forms enamines)



Lesson 3 - Hydride Reductions

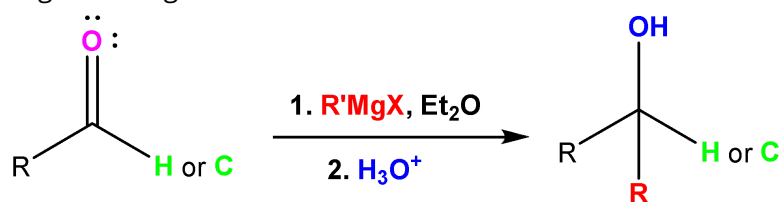


LiAlH₄ is a powerful source of hydride, can also reduce esters

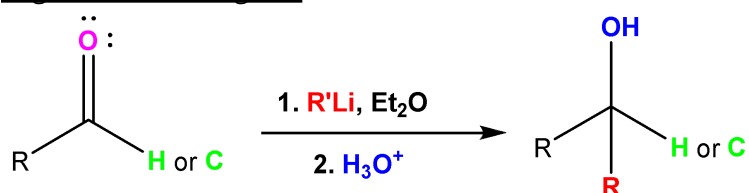


Lesson 4 - Grignards and Organolithium Reactions with Carbonyls

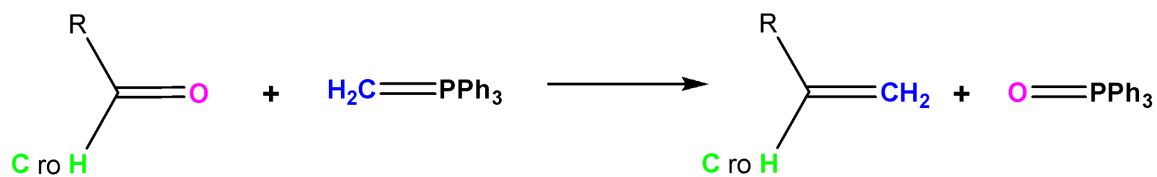
Grignard Reagents



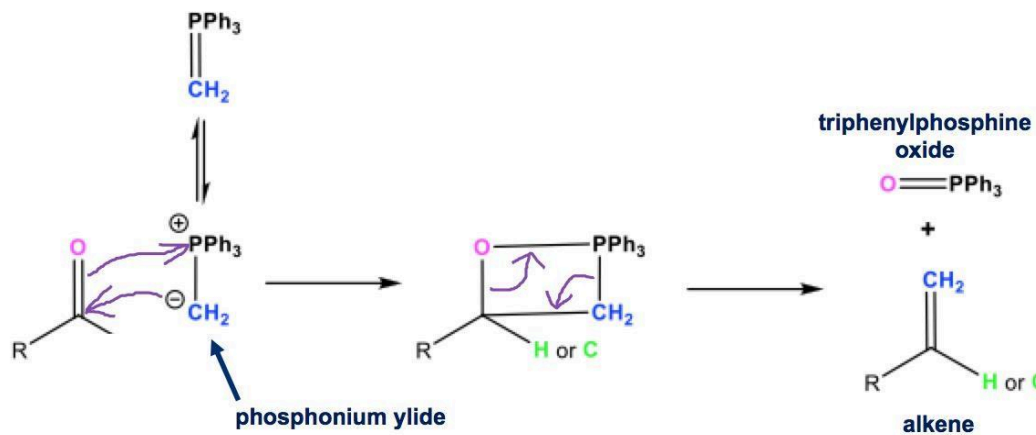
Organolithium Reagents



Lesson 5 - Wittig Reaction

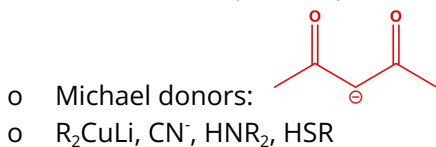


Mechanism of Wittig Reaction

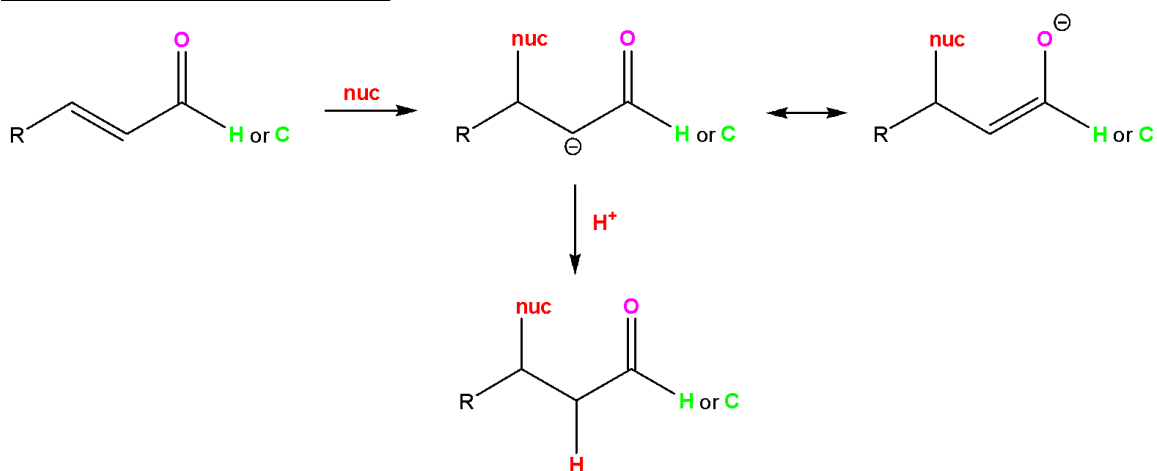


Lesson 6 - Michael Additions (1,4-Addition)

- Add a nucleophile to the β carbon instead of the carbonyl carbon
- Need a weaker nucleophile to prefer the 1,4-addition



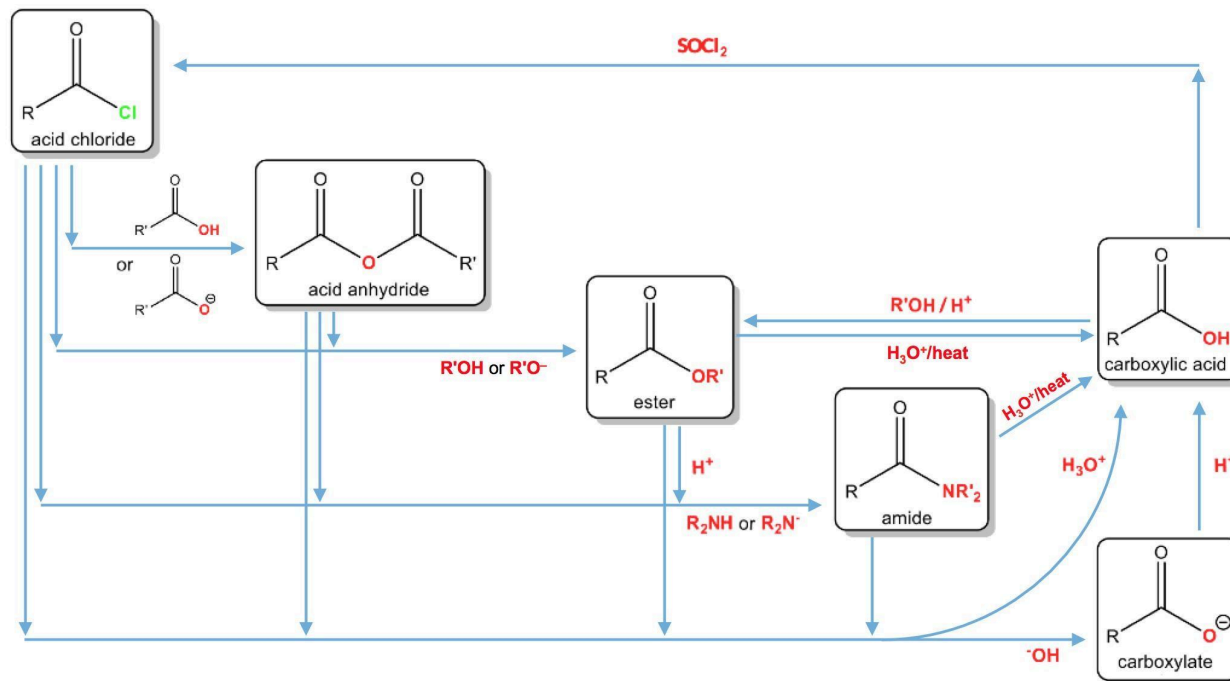
Mechanism of Michael Addition



Ch. 13: Carboxylic Acids and Acid Derivatives

Lesson 1 – Interconversion of Acid Derivatives

Nucleophilic Acyl Substitution



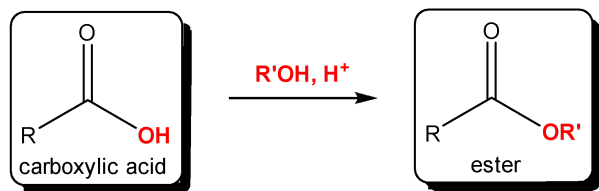
Reactivity = acid chloride > anhydrides > esters/carboxylic acids > amides > carboxylates

Lesson 2 – Physical Properties of Carboxylic Acids

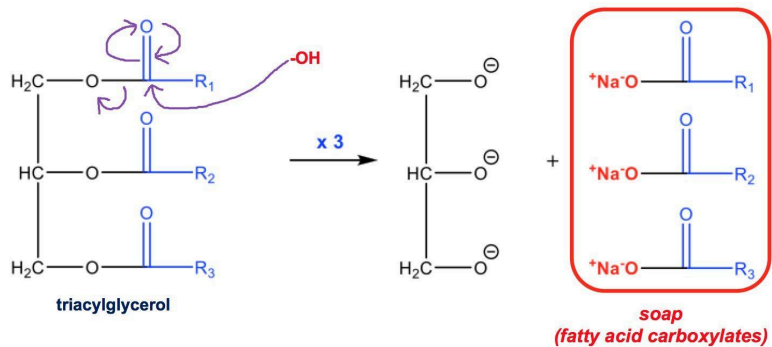
- Carboxylic acids have a higher boiling point due to dimerization and H-bonding

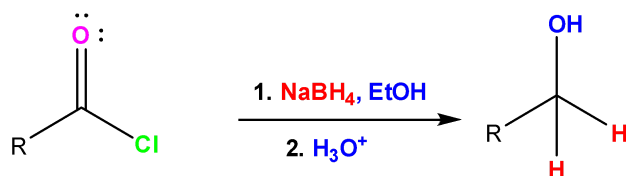
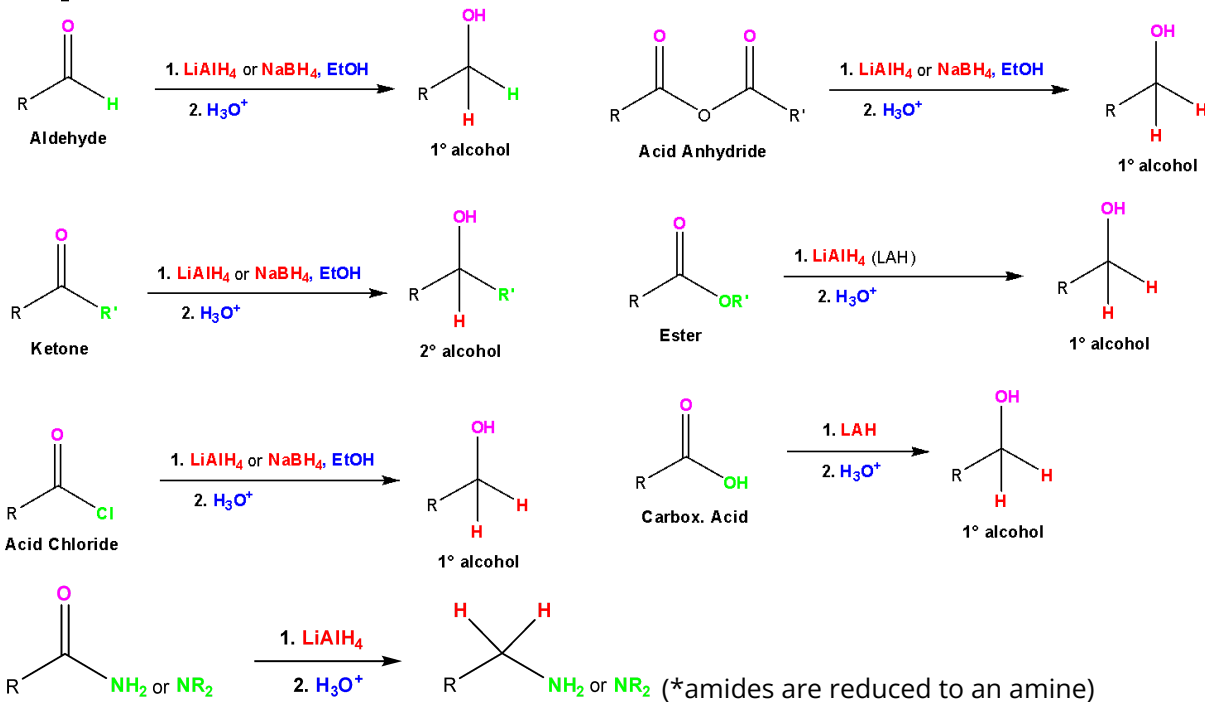
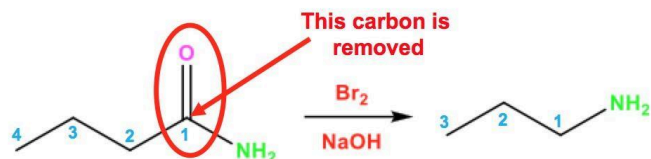
Lesson 3 - Fischer Esterification and Saponification

Fischer Esterification



Saponification

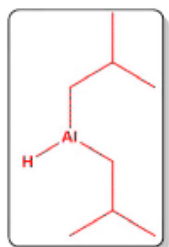
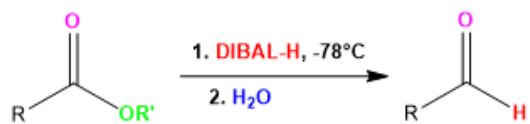


Lesson 4 – Hydride Reductions of Acid DerivativesNaBH₄ reduces ketones, aldehydes, and acid chlorides, and acid anhydridesLiAlH₄ reduces ketones, aldehydes, acid chlorides, esters, carboxylic acids, amides, etc.**Hofmann Rearrangement**

- Turns amides into amines and also removes 1 carbon

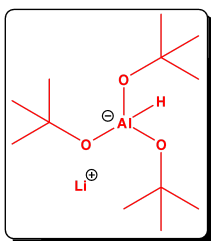
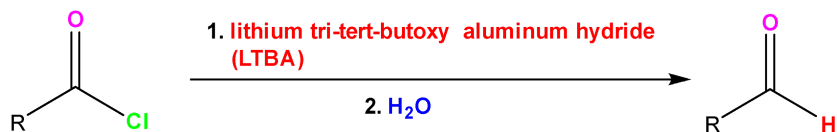
Lesson 5 - Mild Hydride Reductions

DIBAL-H reduces esters to aldehydes



DIBAL-H

LTBA (Lithium tri-*t*-butoxy aluminum hydride) reduces acid chlorides to aldehydes

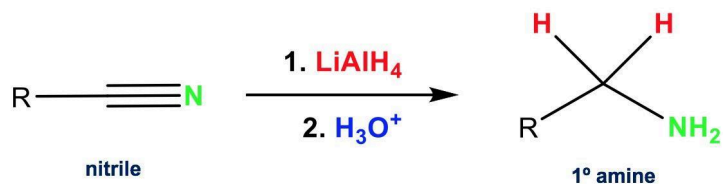


Structure of LTBA

Lesson 6 - LAH and Grignards with Nitriles and Carboxylic Acid Derivatives

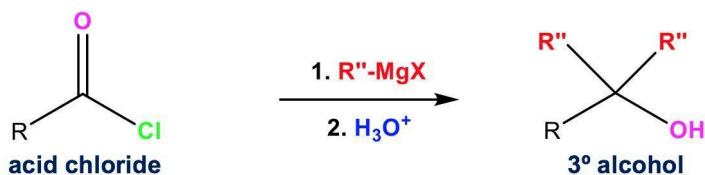
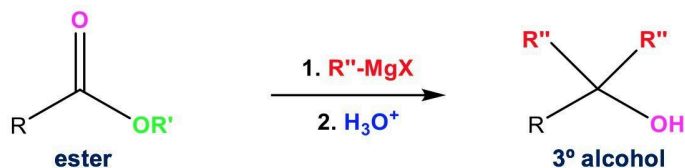
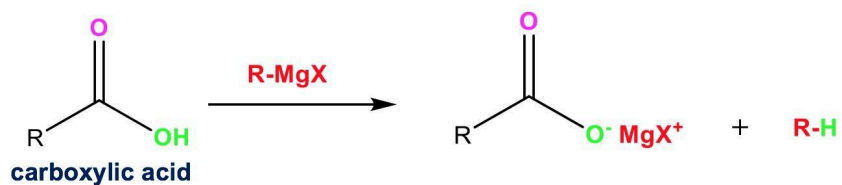
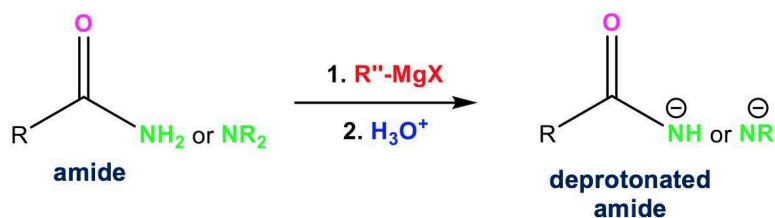
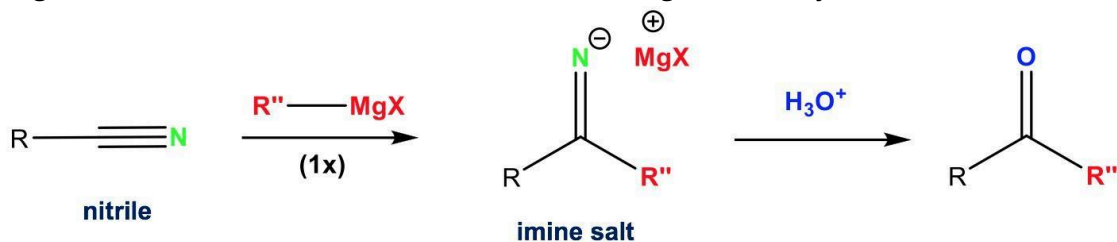
Adding LAH to Nitriles

Reduce nitrile to 1° amine



Lesson 6 - LAH and Grignards with Nitriles and Carboxylic Acid Derivatives (Continued)Reacting Esters and Acid Chlorides with Grignards

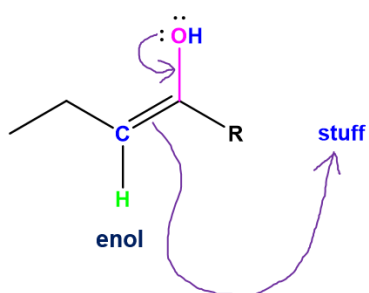
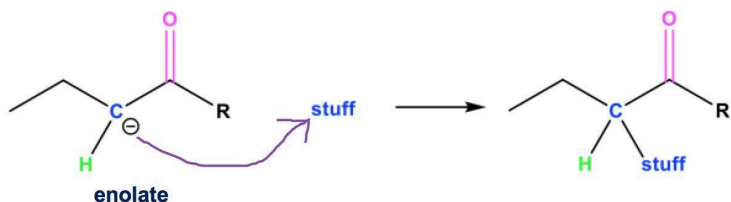
Add the R'' group twice to form 3° alcohol

Reacting Carboxylic Acids with GrignardsGrignard reacts with acidic hydrogen, forms a carboxylate. ***Does not add R'' group**Reacting Amides with GrignardsGrignard reacts with hydrogen on amide, forming deprotonated amide. ***Does not add R'' group**Reacting Nitriles with GrignardsGrignard reacts with nitrile to form **ketone**. Does not go all the way to a 3° alcohol.

Ch. 14: Alpha Substitution Reactions of Carbonyls

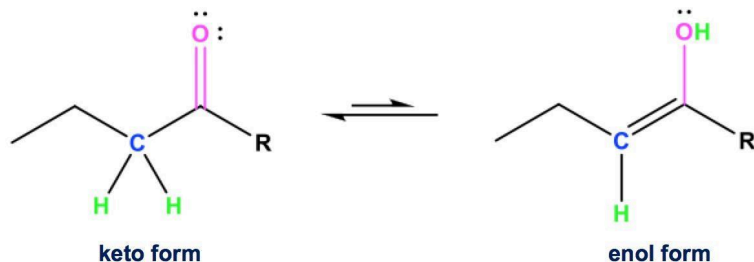
Lesson 1 – Alpha Substitution Reactions

Enolates and Enols



- **Enolates** are better nucleophiles because they have a negative charge

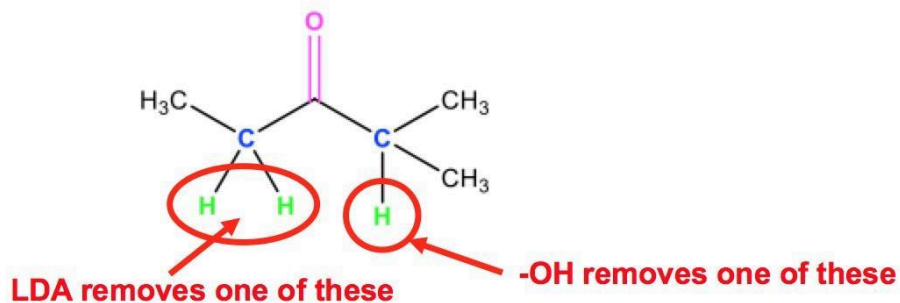
Keto-enol tautomerization



How to deprotonate α -hydrogen

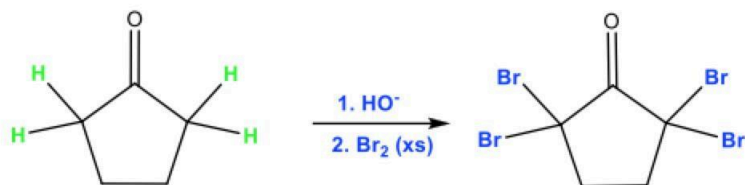
More substituted α -hydrogen gets deprotonated with typical base (-OH)

Less substituted α -hydrogen gets deprotonated with **LDA**

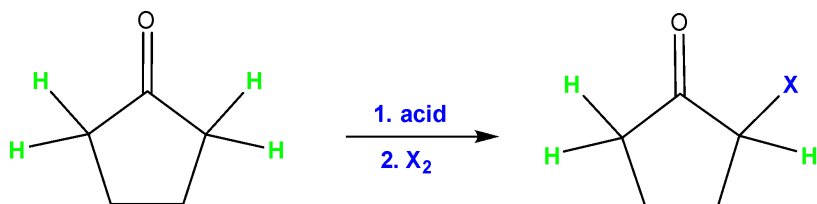


Lesson 2 - Alpha Halogenation and Haloform Reaction

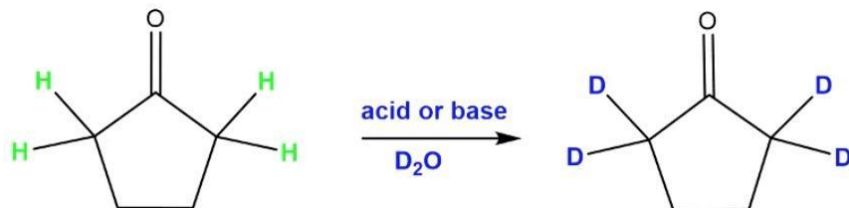
Base-promoted Alpha Halogenation (all alpha hydrogens replaced with halogen)



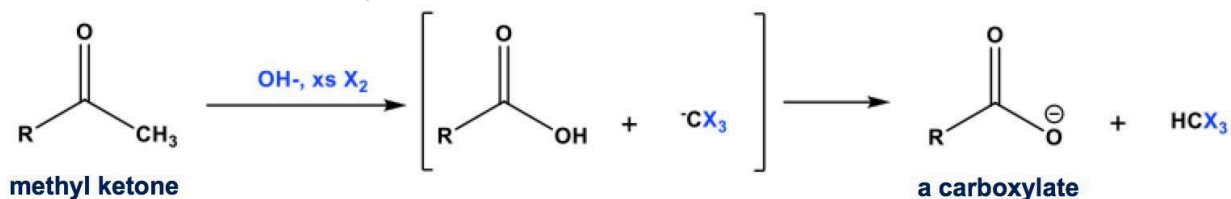
Acid-catalyzed Alpha Halogenation (only one alpha hydrogen replaced with halogen)



Alpha-Deuteration (all alpha hydrogens replaced with D atom regardless of acidic or basic)

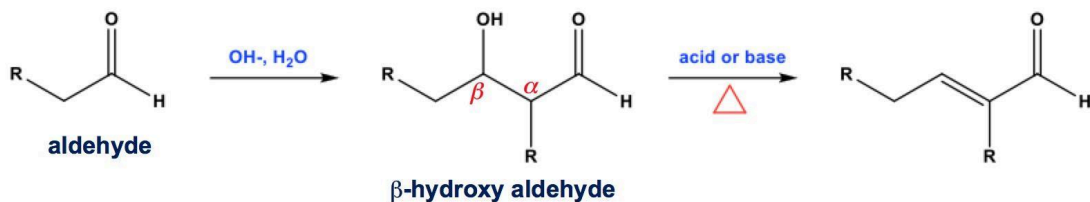


Haloform Reaction (need a CH₃)

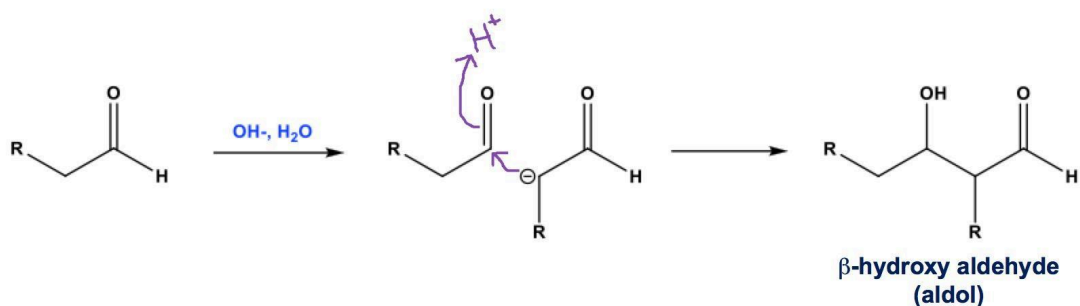


- Produces a yellow precipitate if methyl ketone is present – useful lab test

Lesson 3 - Aldol Condensation

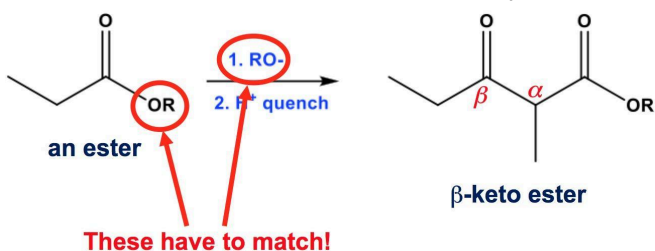


Aldol Shortcut

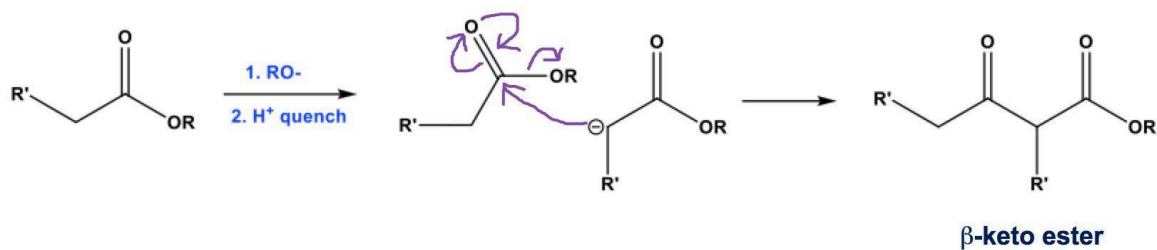


Lesson 4 - Claisen Condensation

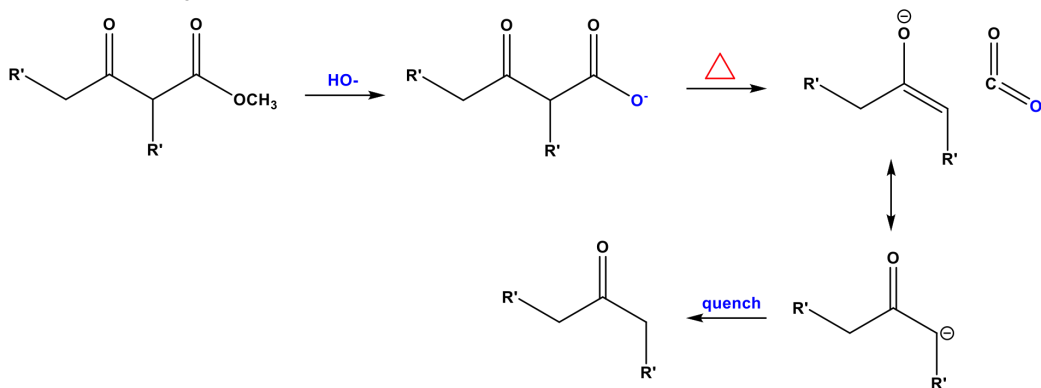
- Just like Aldol Condensation, except with esters. **Enolate** adds to an ester.



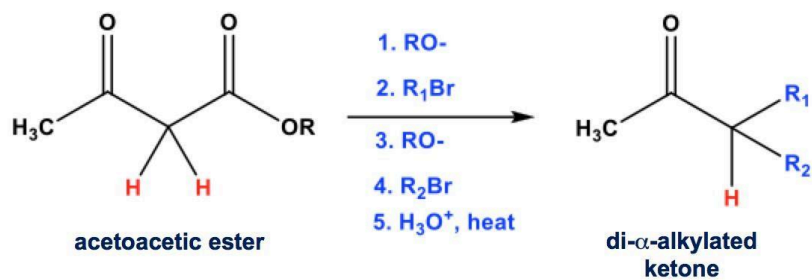
Claisen Shortcut



Beta-Decarboxylation



Lesson 5 – Acetoacetic Ester Synthesis



Lesson 6 – Malonic Ester Synthesis

